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Integrating spatial structure into landslide susceptibility modeling using Gaussian-Based models

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Abstract

Landslide susceptibility assessment refers to a collection of methods aimed at predicting where landslides are likely to occur. Yet, the data used in these models are mostly prone to spatial heterogeneity and dependence, which, if ignored, can lead to biases and inaccurate estimates. In this study, we go beyond the traditional landslide susceptibility framework and use spatial regression methods to estimate landslide frequency in 533 catchments in the Colombian Andes, based on predictors of rainfall, morphometry, geology, and land cover. Moran scatter plots and Lagrange Multiplier tests show that these predictors are spatially autocorrelated and clustered. While landslide occurrence is inherently discrete, we systematically compare an Ordinary Least Squares (OLS) regression against several Gaussian-based spatial models to assess their ability to handle spatial structures in the data. We employ Geographically Weighted Regression (GWR) and Spatial Regimes to address spatial heterogeneity, and Simultaneous Autoregressive (SAR) models to account for spatial dependence. Our results show that accounting for these spatial properties

significantly improves model fit and provides deeper geomorphic insights. GWR and Spatial Regimes reveal distinct regions where relief and elevation exert varying influences, while the superior performance of the Spatial Error Model highlights the impact of spatially autocorrelated, unobserved factors. We conclude that, despite the limitations of a Gaussian framework for count data, integrating spatial dependence and heterogeneity is essential for generating more statistically sound and geographically realistic landslide susceptibility models.

Keywords: landslide; susceptibility; spatial dependency; spatial heterogeneity, Gaussian models, Colombia.

1. Introduction

Landslide susceptibility studies rely on a suite of methods that estimate the likelihood that a slope failure occurs in a particular part of a landscape (Brabb, 1984; Corominas et al., 2014; Fell et al., 2008). Such estimates hinge on statistical models that use a combination of selected geological or environmental predictor variables. The local values of these are routinely obtained for discrete terrain mapping units, which may include regular and arbitrary areas such as grid cells (Ba et al., 2018); irregular but geomorphologically meaningful units such as slope units (Alvioli et al., 2016; Bai et al., 2015); or spatial units that do not necessarily reflect natural boundaries, such as municipality boundaries (Reichenbach et al., 2018).

The response variable in these models is the likelihood of landslide occurrence, which can be expressed in several ways depending on the modeling approach. In most data-driven models, the response variable used for model training is derived from a landslide inventory and is binary, indicating presence or absence of mapped landslides within each mapping unit. Logistic regression models, for instance, predict the probability of classifying a given unit as part of a previously mapped landslide. Other approaches, such as Weights-of-Evidence models (Bonham-Carter et al., 1989), have a continuous response variable that reflects the relative contribution of each predictor.

In physically-based models, a training dataset is not required; instead, the response variable is often a continuous and strictly positive value such as the Factor of Safety (Durmaz et al., 2023), which quantifies slope stability. Other ways to express landslide susceptibility include discrete count response variables, such as the number of landslides per unit area modeled using Poisson or negative binomial distributions, or proxies such as landslide density, total affected area, or the proportion of terrain impacted (Reichenbach et al., 2018; Fell et al., 2008). These alternative representations allow for greater flexibility in statistical modeling, enabling the use of different distributions and offering complementary information on landslide hazard in terms of intensity or frequency.

Although landslide susceptibility estimation is a spatial modeling problem, most studies assumed that predictors are mutually independent and spatially homogeneous (stationary) (Xiao et al., 2021). This assumption is in conflict with the geospatial nature of the data. Landslide-controlling factors, and the resulting landslide inventories, have some spatial dependence, where nearby landslides are more similar than distant ones, and spatial heterogeneity, where predictors vary across space (Fang et al., 2024; Smith et al., 2023; Liu et al., 2024; Zhang et al., 2023; Quevedo et al., 2022; Lombardo et al., 2020, Anselin, 1988). This spatial structure violates basic assumptions of classical regression models, such as the independence and constant variance of residuals, and negligible multicollinearity among predictors (Cressie, 2015; Ripley, 1988; Anselin, 1988). Violating these assumptions means that some degree of spatial structure remains in the residuals of these models such that part of the variance and its source can be misestimated (Opitz et al., 2022; Rey et al., 2023). Spatial heterogeneity can lower the validity of model estimates, bias parameter estimates, create misleading significance levels, and poor forecasts (Li et al., 2022). Similarly, spatial dependence may produce misleading significance levels of T-tests, and inaccurate R^2-values due to biased estimates of the error variance (Anselin, 1990).

Despite these caveats, it remains a widespread practice to ignore spatial structure when estimating landslide susceptibility (Fotheringham et al., 2000; Schabenberger & Pierce, 2001). Meanwhile, spatial models have become popular for addressing dependency and heterogeneity in fields such as ecology and disease mapping (Bivand et al., 2021), under the term Bayesian hierarchical regression models (Haining & Li, 2020) and spatial econometrics (Anselin, 1988; Arbia, 2016). These techniques capture spatial relationships and local variations that traditional models fail to account for, but have hardly been explored in landslide susceptibility studies (Di Napoli et al., 2023; Fang et al., 2024; Liu et al., 2024), partly because their complexity (Bryce et al., 2024).

Spatial modeling generally follows one of two main strategies. One is to treat data as if they were obtained from a continuous field, and appropriate models include Gaussian Processes (Rue & Held, 2005), for example kriging (Cressie, 2015). The spatial structure of the data is encoded by a covariance matrix based on the pairwise distances between the observations. Thus, Gaussian Processes are computationally demanding for large datasets with commensurately large covariance matrices (Banerjee et al., 2008), likely limiting this method for landslide susceptibility (Bryce et al., 2022).

The second strategy considers data in discrete spatial units and includes two main families of models: autoregressive models and spatially varying coefficient (SVC) models. Autoregressive models (Anselin, 1996; Yadav et al., 2023) introduce spatial structure through a covariance function that links each observation with its neighbors, rather than with all other observations as in Gaussian Process models (Fischer & Wang, 2011; Fotheringham et al., 2000). This approach allows for several ways of defining neighborhood relationships among discrete data (Getis, 2009) and is computationally efficient, straightforward to implement, and easily interpretable in relation to nearby units. In contrast, SVC models allow regression coefficients to vary across space, either at the level of each individual unit or over spatially homogeneous regions.

This study focuses on explicitly integrating and evaluating a Gaussian-based spatial regression framework to model landslide susceptibility in the Colombian Andes. First, we address a common methodological pitfall in landslide susceptibility by systematically quantifying the extent of misestimation that occurs when using a non-spatial Ordinary Least Squares (OLS) model. While most studies favor logistic regression or machine learning classifiers, we compare OLS against a suite of spatial models specifically designed to manage spatial heterogeneity and dependence, highlighting the critical information lost when spatial structures are neglected. Second, we use this framework as a diagnostic tool to investigate how accounting for spatial structure provides a more nuanced, geomorphologically-interpretable understanding of landslide controls. We argue that these models are not just statistical refinements, but essential tools for revealing how the influence of predictors such as relief or elevation varies across the landscape, an insight obscured by traditional, spatially-static models.

2. Spatial regression models for discrete data

Spatial models consider two types of spatial structure in data, i.e. heterogeneity and dependence (Anselin, 1988). These terms are also known as non-stationarity and autocorrelation in time series analysis, respectively. Unlike time series that are one-dimensionial, spatial data can involve more dimensions. Figure 1 summarizes various spatial regression models and their relationships. These include models designed for continuous spatial fields, such as Gaussian Processes (GP), and models for spatially discrete data, which can be further categorized into autoregressive (AR) models that address spatial dependence, and spatially varying coefficient (SVC) models that capture spatial heterogeneity.

2.1 Spatial heterogeneity

Spatial heterogeneity refers to domain effects on the sample units, where the local mean as a first-order statistical moment varies from place to place (Zhang et al., 2023; Wang et al., 2022). Spatial heterogeneity in landslide susceptibility invites using two types of SVC models, i.e. Geographically Weighted Regression and Spatial Regimes.

The most common SVC model is Geographically Weighted Regression (GWR) (Brunsdon et al., 1996; Fotheringham et al., 2009). In GWR models, the relationship between predictors and the response varies continuously across geographic space using a local weighting matrix (Peng & Inoue, 2024). GWR enables us to examine how geomorphic drivers of landsliding vary spatially, testing the hypothesis that the explanatory power of each predictor is modulated by local geographic settings rather than being a universal constant. For instance, the effect of slope on landslide occurrence might be very strong in coherent bedrock but weak in unconsolidated sediments. GWR produces a continuous surface of coefficient estimates of where and how these relationships change. This method captures spatial variation by assigning greater weight to nearby observations using a kernel function, such as Gaussian, exponential, or bi-square (Fotheringham et al., 2000; Opitz et al., 2022). This kernel function needs a specified bandwidth that defines the neighborhood of influence around each point. The bandwidth can be either fixed in terms of a geographic distance, or adaptive by adjusting to a set number of neighboring observations (Brunsdon et al., 1996). The general formulation of a GWR model for a given location i is:

$$y_i = \beta_{i0} + \sum_{k=i}^{k} \beta_{ik}\, x_{ik} + \varepsilon_i$$

where y_i is the value of the response variable; β_{i0} is the intercept; β_{ik} is the kth coefficient, x_{ik} is the kth predictor variable; k is the number of predictors, and ε_i is the random error term.

Spatial Regime (SR) models belong to the family of multilevel models, where observations are assigned to distinct groups or regimes that represent conceptually defined regions (Anselin, 1996). These regimes function as levels in the multilevel framework, allowing for the examination of structural differences between regions. In SR models, regression coefficients are estimated for each regime, capturing how relationships between predictors and the dependent variable may differ across these predefined spatial zones. The spatial pattern in these models is determined by the geographical or contextual delineation of regimes, which can be based on physical boundaries (e.g., basins or spatial clusters), land use classes, or other spatial criteria (Piras & Sarrias, 2023). Rather than modeling continuous spatial variation, SR models focus on within-regime variations while allowing comparisons of parameter differences across regimes. This approach retains the multilevel character of the model, as it accounts for both within-regime effects (individual-level variation) and between-regime differences (group-level variation). SR models can also include spatially fixed effects, where each regime has its own set of coefficients.

2.2 Spatial dependency

Spatial dependence means that adjacent observations are more similar or correlated than those farther apart (Fletcher & Fortin, 2018), estimated by the covariance as a second-order moment (Wickham, 2008). This spatial dependence is modeled by incorporating a spatial lag as a predictor, which represents a weighted average of neighboring values. The spatial lag is computed using a neighborhood connectivity matrix **W** that defines the spatial relationships between units (Anselin, 1996; Rey et al., 2023).

Both Conditional Autoregressive (CAR) and Simultaneous Autoregressive (SAR) models can handle spatial dependency in discrete areal data by relying on neighborhood information (Wall, 2004; Jaya & Chadidjah, 2021; Cressie, 2015). SAR models explain relationships among response variables at all locations simultaneously (Ver Hoef et al., 2018; Jaya & Chadidjah, 2021), whereas CAR models estimate local response conditional on values at neighboring locations (Anselin, 1988). CAR models are defined within the framework of Bayesian Multilevel models, where spatial dependence is introduced as a structured random effect through a Markov Random Field.

In SAR models, the treatment of spatial dependence across all locations gives rise to what is known as a spillover effect—that is, a change in a predictor or in the response variable at one location can propagate and indirectly influence values at other locations, including those not directly connected through the neighborhood matrix (LeSage, 2014). The spillover effect quantifies how a change in a predictor at one location (i) can influence the response variable at other locations (j) (LeSage, 2014; Bivand, 2022). While this effect is mathematically well-defined within the spatial weighting matrix, its physical interpretation in a landslide context must be handled with caution. This property makes SAR models suited for capturing spatial diffusion processes and global interaction effects.

Neighborhood connectivity matrices can be obtained in several ways, i.e. by (i) *p*-order binary contiguity matrices that consider *p*-order neighbors (where *p* is a positive integer representing first-order neighbors *n*=1, second-order neighbors *n*=2, and so on, up to *p*=*n* levels); (ii) inverse distance matrices or exponential distance decay matrices; or (iii) *q*-nearest neighbor matrices, where *q* is a positive integer representing the number of neighbors considered (Getis & Aldstadt, 2004; Getis, 2009; Stakhovych & Bijmolt, 2009).

In SAR models, spatial lag can be introduced in different model terms, leading to three main types of spatial dependency: (i) exogenous autocorrelation, where the spatial lag is applied to the predictor variables (**WX**); (ii) endogenous autocorrelation, where the spatial lag affects the response variable (**W**y); and (iii) autocorrelation in the error term, where spatial structure is in the residuals (**W**μ), leaving the predictors uncorrelated with the composite error (Anselin, 1988). The General Nesting Spatial (GNS) model—also known as SARAR(1,1) or the Cliff-Ord spatial model—encompasses all three forms of spatial dependence (LeSage & Pace, 2009; Kelejian & Prucha, 2007; Cliff & Ord, 1981):

$$y = X\beta + WX + \rho W y + \lambda W \mu + \epsilon$$

where the positive $n\times n$ spatial weight matrix $\mathbf{W}$ describes the structure of dependence between n observations in a $n\times 1$ vector y, in which each element y_i represents the response variable in spatial unit i; β is a $k\times 1$ vector of k coefficients for the predictors in design matrix $\mathbf{X}$ with $n\times k$ dimensions; ρ is the coefficient of the spatially lagged dependent variable; λ is the coefficient in a spatial autoregressive error; μ and ε are the spatial and local error terms of the model. Here, μ follows a first-order spatial autoregressive process, whereas ε is a $n\times 1$ vector of independently and identically distributed errors with zero mean and variance σ^2. The term $\rho\mathbf{W}$ captures endogenous interaction effects of the response variable, while $\mathbf{WX}$ models the exogenous interaction effects among the predictors; the term $\lambda\mathbf{W}$ deals with the interaction effects among the residuals.

Among the simplified variants of the SAR model, the Spatially-lagged y Model (SLy), where $\lambda=0$ and $\mathbf{WX}=0$, thus only featuring the spatially lagged dependent variable, $\rho\mathbf{W}y$. In the case that $\rho=0$ and $\mathbf{WX}=0$, the Spatial Error Model (SEM) retains the error terms $\lambda\mathbf{W}\mu$, while the Spatial Lagged X Model (SLX) comprises the spatial lags of the predictors, $\mathbf{WX}$, when $\lambda=0$ and $\rho=0$ (Elhorst, 2013). Further variants include the Kelejan-Prucha or Spatial Autoregressive Combined Model (SAC) that includes autocorrelation in $\rho\mathbf{W}y$ and the $\lambda\mathbf{W}\mu$; the Spatial Durbin Model (SDM) combines an autoregressive dependent variable $\rho\mathbf{W}y$ and spatially lagged predictors $\mathbf{WX}$; and the Spatial Durbin Error Model (SDEM) has a spatial error term $\lambda\mathbf{W}\mu$ combined with spatially lagged predictors $\mathbf{WX}$ (Elhorst, 2022; Anselin, 2022); (Fig. 1).

In models with endogenous dependence (e.g., SLy, SDM), the spillover effect is global: a change in one catchment propagates through the system-wide connectivity defined by the spatial weights matrix ($\mathbf{W}$). Due to the simultaneous feedback nature of the autoregressive term (ρ), an impact in one unit simultaneous feedback its neighbors, then to the neighbors of those neighbors, eventually influencing the entire study area. This global spillover implies a form of systemic propagation that is physically difficult to justify for landslides, as they are discrete slope failures rather than a contagious process. Instead, this mathematical dependence suggests that the spatial structure is likely driven by the predictors; the environmental drivers within each mapping unit exhibit strong spatial autocorrelation, which in turn conditions the distribution of landslide counts. Therefore, the model captures the spatial organization of the susceptibility factors rather than a direct physical interaction between individual landslide events. Conversely, in models with only exogenous dependence (e.g., SLX, SDEM), the spillover is restricted to the immediate spatial neighborhood defined by **W**. This framework is geomorphologically more plausible as it represents localized interactions; for example, it accounts for how intense rainfall or specific geological conditions in an adjacent catchment—rather than across the entire study area—may influence the landslide count within a given unit. In this sense, the model captures the 'neighborhood effect' of environmental drivers, reflecting the spatial scales at which triggering factors typically operate. The SEM model, by contrast, confines all spatial dependence to the error term and assumes no spillover in the response variable. Therefore, selecting among these models is not just a statistical choice; it is a decision about the assumed spatial nature of the underlying environmental and landscape processes, whether they are localized, systemic, or driven by latent, i.e. unobserved, factors (Elhorst, 2013).

2.3 Spatial Model Tests

The first step in exploring the spatial structure of predictors is to test for autocorrelation. The Moran scatter plot and its associated Moran's I statistic, and the Lagrange Multiplier (LM) test for omitted spatial effects (Anselin, 1996). The scatter plot relates each standardised observation to the spatially lagged mean of its neighbours (Anselin, 1988); the slope of a least-squares fit is Moran's I,

$$I = n \frac{\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} \, (y_i - \bar{y}) \left(y_j - \bar{y} \right)}{\left(\sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} \right) \left(\sum_{i=1}^{n} (y_i - \bar{y})^2 \right)}$$

where n is the number of observations, y_i is the value of the response variable at location i, $\bar{y}$ is its mean, and w_{ij} are elements of the row-standardised spatial-weight matrix **W**. Moran's I ranges from -1 (perfect dispersion) to +1 (perfect clustering), with zero indicating spatial randomness. Because global measures can mask local spatial patterns, we also compute Local Indicators of Spatial Autocorrelation (LISA) (Anselin, 1996). LISA decomposes Moran's *I* into location-specific contributions and obtains pseudo-*p* values via permutation, thus pinpointing statistically significant "hot spots", "cold spots", and spatial outliers.

The LM test is used to decide on an appropriate spatial regression model and to estimate spatial autocorrelation in both the response variable and its residuals. The test compares a ordinary least squares model with unrestricted models that have spatial terms. The spatial term can either be a spatially lagged dependent variable or a spatially lagged error term. The LM statistic is calculated as the difference between the log-likelihoods of the restricted and unrestricted models, following a χ^2 distribution with one degree of freedom under the null hypothesis of no spatial dependence. If the test statistic exceeds the critical value, the null hypothesis of spatial independence is rejected.

3. Data

3.1 Study area

Our study area is located north of 5°N and encompasses the Western and Central Cordilleras of the Colombian Andes, separated by the Cauca canyon. The region is bound by the Atrato River to the west and the Magdalena River to the east. Covering 50,000 km², the area has 533 catchments of medium order within the Atrato (25% of catchments), Cauca (50%), and Magdalena basins (25%). Roughly 73% of these catchments drain areas less than 100 km²; the median area is 48 km².

3.2 Response variable

We used a catalogue of 13,777 shallow and deep-seated landslides that occurred between 1970 and 2023 (Aristizábal & Korup, 2025) (Fig. 2), detected manually using high-resolution (<1 m) true-color optical images from Google Earth™. While this inventory provides a comprehensive regional dataset, we acknowledge several uncertainties due to its compilation from optical imagery (Google Earth™). The manual detection process is subject to several potential sources of spatial bias, such as variable image resolution and cloud cover across the study area, which may lead to under-reporting in certain regions. This inventory is thus considered a representative, albeit incomplete, sample of landslide occurrence.

We use landslide abundance—the number of landslides per spatial unit—as the response variable. Instead of merely classifying a catchment as susceptible or not, modeling landslide abundance allows us to differentiate between areas based on their relative levels of observed landslide occurrence, offering a more comprehensive and informative view of landslide susceptibility. In our case, the empirical distribution of landslide abundance in the Colombian Andes is characterized by a high frequency of zero counts and a strong positive skew. Our modeling approach is based on regression techniques that assume a Gaussian distribution, which is not not applicable to count data. To adapt our data to this framework, we transformed the response variable using a logarithmic function ($\log(Y + 1)$); adding 1 avoids zero landslide number values, whose logarithm would produce undefined results. This step transforms the skewed, discrete data into an approximately continuous variable, stabilizes the variance, and helps the subsequent model's residuals approach a Gaussian distribution (Figure 3).

3.3 Predictor variables

Initially, we selected nine geomorphological, geological, and climatic candidate predictors to model landslide abundance at the catchment scale, drawing on their reported relevance in earlier studies (Reichenbach et al., 2018; Van Westen et al., 2008). These predictors include: the hypsometric integral (H_i), which reflects the stage of landscape evolution and potential for landsliding; mean elevation (E) and mean slope (S), both of which influence gravitational driving forces; mean local relief (H), indicating terrain ruggedness; lineament density (Ld) representing structural controls such as faults and fractures that can weaken slopes; mean annual rainfall (P), a key factor in weathering processes; and land use cover (Lc) and geology (G), which reflect soil properties, vegetation cover, and material strength. We also included the number of days with daily rainfall exceeding 20 mm (P_{20}) because short-term intense rainfall events are commonly associated with landslide initiation in the Colombian Andes (Aristizábal & Sánchez, 2020).

To align all predictors with the catchment-level response variable, we aggregated their values within each catchment: continuous variables were averaged, lineament density was calculated, and categorical variables were represented by their modal class. All terrain parameters were calculated using the 12.5-m digital elevation model (DEM) from the Advanced Land Observing Satellite-Phased Array-Type L-Band Synthetic Aperture Radar (ALOS-PALSAR) (Logan & Karr, 2014). Lineament density (*Ld*) was estimated using the LINE algorithm from the image processing software CATALYST, based on SRTM data (Villalta & Smith, 2002), and for eight hillshade directions at 45° bins of azimuth with a sun elevation angle of 45°. We derived rainfall information from the Climate Hazard Group InfraRed Precipitation with Station Data (CHIRPS) (Funk et al., 2015), version 2.0, at 5-km resolution from 1981 to 2023, using Google Earth Engine (GEE). We generated the land cover map using 10-m Sentinel-2 imagery from the Copernicus Programme. We selected images from 2018 to 2023 with <10% cloud coverage, removed shadow effects, and created layers of the Normalized Difference Vegetation Index (NDVI) (Kriegler, 1969), Normalized Difference Built-up Index (NDBI) (Zha et al., 2003), Modified Normalized Difference Water Index (MNDWI) (Xu, 2006), and Bare Soil Index (BSI) (Li et al., 2014). We performed land cover classification using a random forest algorithm implemented with the GEE Classifier package (Breiman, 2001).

To implement SR models, it is necessary to establish spatial regimes or homogeneous regions. We tested two regimes: the first regime reflects the major drainage basins in our study area, i.e. Atrato, Cauca, and Magdalena. However, this hydrological segmentation may not necessarily fully explain the abundance of landslides. Hence, for the second regime we used spatial clustering on morphometric parameters to identify homogeneous regions that might explain the spatial patterns of landslide occurrence. We used Agglomerative Clustering using K-nearest neighbors for K ∈ {2, ..., 6}, using the Python Scikit-Learn library, and guided by the elbow method and Silhouette coefficient (Pedregosa et al., 2011). The elbow method helps identify the optimal number of clusters by plotting the Within-Cluster Sum of Squares (WCSS) against K to detect an inflection point (Thorndike, 1953). The Silhouette coefficient (Sc) assesses cohesion within, and separation between, clusters, and ranges from −1 (poor clustering) to +1 (strong clustering) (Rousseeuw, 1987). For handling spatial heterogeneity, we used the Python MGWR library by Oshan et al. (2019), and the PySAL library by Rey & Anselin (2007). For estimating spatial dependency, we used the R libraries spdep and spatialreg by Bivand (2022). The impacts function from the R package spatialreg helped to estimate direct and indirect impacts of predictors.

3.4 Model performance

To compare the performance of all selected models, we use Akaike's Information Criterion (AIC) along with the adjusted R^2 or the Nagelkerke R^2. The AIC evaluates model performance by balancing goodness of fit against the number of estimated parameters. The choice between adjusted R^2 and Nagelkerke R^2 depends on the model type. For the non-spatial OLS model, we report the adjusted R^2 to measure the proportion of variance explained by the predictors while accounting for the degrees of freedom. For models estimated via Maximum Likelihood, we use the Nagelkerke pseudo- R^2 (Nagelkerke, 1991), which provides a comparable measure of explanatory power by normalizing the likelihood ratio. These metrics allow us to evaluate the goodness of fit across different estimation frameworks while penalizing the inclusion of unnecessary parameters. In contrast, for models estimated through maximum likelihood estimation (MLE)—such as SAR and SR models—we use the Nagelkerke R^2, which expresses the improvement of model likelihood relative to a null model, and is scaled between 0 and 1. We include these measures to provide an intuitive understanding of model performance, with AIC remaining the primary criterion for selecting the most parsimonious model.

4. Results

4.1 Variable selection

To reduce multicollinearity and reduce the dimensionality of the environmental dataset, we first conducted a Principal Component Analysis (PCA) on the set of standardized predictor variables. We retained components with eigenvalues greater than 1 (Kaiser's criterion (Kaiser, 1960)) and used the loading scores to group collinear variables. Variables with high loadings (>0.6) on the same principal component were considered collinear and evaluated for exclusion or aggregation.

The first two principal components explained 70% of the total variance—with PC1 accounting for 49% and PC2 for 21%. Figure 4 illustrates the PCA results, highlighting four groups of collinear predictors: (i) lineament density (Ld), mean local relief (H) and mean slope (S); (ii) hypsometric integral (H_i) and mean elevation (E); (iii) mean annual rainfall (P) and days with rainfall > 20 mm (P_{20}); and (iv) Area (A). Predictors Lc, G and Ld have coefficient estimates that may hardly differ from zero. Conversely, Ld shows high correlation values with H (0.81) and S (0.91), while H and S correlate (0.96). Similarly, the two rainfall predictors, P and P_{20}, are highly correlated (0.98), and negatively correlated with E (-0.6). Although, H_i shows low correlation values with the other predictors, the coefficients were very low and had no statistical significance.

Within each group we selected a single representative predictor while discarding the remaining collinear predictors. Categorical variables (land-use cover Lc, geology G) were excluded from the PCA, and we retained them in their modal form because of their clear interpretative value. This screening step provides a parsimonious, interpretable set of candidate predictors for the subsequent regression analysis.

4.2 Exploratory spatial structure

To define the spatial weight matrix, we used the *k*–nearest-neighbour (k-NN) approach, which guarantees that every catchment has at least one neighbor and avoids isolated units with no spatial links. Moreover, it allows weighting neighbor influence as a function of distance, assigning greater weight to closer catchments. We evaluated matrices with $2 < k < 9$, and used Moran's I and Local Indicators of Spatial Association (LISA) to identify the value of k that maximized both global and local autocorrelation. Results indicated that k=5 provided the highest spatial coherence while keeping **W** sparse. Figure 5 shows the resulting spatial connectivity network, which we use for all subsequent analyses.

Using the k=5 matrix, we computed Moran's I and Local Indicators of Spatial Autocorrelation (LISA) for every morphometric covariate. Slope, elevation, and relief have a strong positive spatial autocorrelation ($I>0.8$), whereas H_i has a moderate structure. The LISA map for slope (Fig. 6) shows statistically significant hot spots (High–High) and cold spots (Low–Low).

To delineate spatial regimes we ran a k-means clustering of the standardised morphometric variables, using the same k=5 weight matrix as spatial constraints. The optimal solution, obtained by minimising the within-cluster sum of squares and maximising the silhouette width, has four clusters (A–D; Figure 7). Cluster A corresponds to the low-lying alluvial terrain along the Magdalena and Cauca basins; cluster B represents the high-relief terrain along the Cauca canyon; cluster C is the low alluvial terrain along the Atrato basin, and cluster D comprises the high-elevation, low-relief surfaces along the axis of the central Cordillera. Both regimes (basin and morphometric clusters) are used in the spatial models.

4.3 Standard linear regression model (non-spatial)

Using the reduced predictor set obtained, we fitted several multivariate, non-spatial regression models with OLS estimation. Table 1 lists the parameter estimates for the standard linear regression models based on various combinations of standardized predictors. These non-spatial OLS models serve as the benchmark for gauging the benefits of incorporating spatial effects in more sophisticated models.

Table 1: Parameter estimates for non-spatial, standard regression models using standardized predictor variables. Bold values indicate statistically significant parameter estimates at $p<0.05$, with standard errors in parentheses; model performance is expressed by Adjusted R^2 and Akaike Information Criterion (AIC).

Predictor	**OLS1 (all predictors)**	**OLS2**	**OLS3**	**OLS4**	**OLS5** (selected predictors)
Constant	**1.814** (0.074)	**1.840** (0.056)	**1.862** (0.086)	**1.759** (0.076)	**1.840** (0.048)
A	**0.586** (0.050)	-	-	**0.603** (0.051)	**0.557** (0.049)
H_i	0.069 (0.058)	0.104 (0.063)	0.035 (0.067)	-	-
Ld	0.179 (0.150)	**-0.449** (0.142)	-0.024 (0.106)	-	-
P	-0.077 (0.067)	**-0.253** (0.063)	**-0.272** (0.066)	-	-
E	**0.423** (0.074)	-	-	-	**0.514** (0.058)
S	-0.123 (0.298)	**1.112** (0.146)	-	**0.415** (0.191)	-
H	**0.573** (0.204)	-	**0.792** (0.103)	**0.492** (0.182)	**0.546** (0.058)
Lc(grass)	0.202 (0.123)	-	0.197 (0.143)	**0.504** (0.119)	-
G	-0.002 (0.171)	-	-0.242	-0.293	-

Predictor	**OLS1 (all predictors)**	**OLS2**	**OLS3**	**OLS4**	**OLS5** (selected predictors)
(sediment)			(0.194)	(0.173)	
G (volcanic)	-0.131 (0.125)	-	-0.189 (0.145)	-0.135 (0.126)	-
Adj. R^2	0.500	0.319	0.318	0.445	0.497
AIC	1611.636	1768.473	1772.166	1662.796	1608.693

The best OLS model of the ones tested here was that with predictors catchment area A and terrain parameters H and E, and landslide catchment abundance tends to be significantly higher with increasing local relief and higher elevations. This model explains only 50% of the variance in the landslide number per catchment. We examined the model residuals for spatial structure (Figure 8) both globally and within specific spatial regimes. By analyzing the spatial distribution of residuals across basins and morphometric clusters, we sought to identify regions where the model consistently over- or under-predicts landslide abundance. Such spatial variations in residual patterns suggest that the underlying geomorphological or hydrological drivers may operate differently across regimes, indicating that a single global parameters set may not fully capture the local complexities of the landscape.

Residuals are higher in catchments in the Cauca and Magdalena basins, indicating poorer model fit in these regions. Similarly, clusters B and D, where landslides are most abundant, show higher residual variances, suggesting that local terrain variations may not be fully captured by the non-spatial model. The Moran's I for the global residuals of 0.376 ($p = 0.001$) indicates spatial autocorrelation. Similarly, the Lagrange Multiplier (LM) tests yield significant statistics for both the error term (RS_err = 205.64) and the spatial lag of the dependent variable (RS_lag = 182.90). The higher value for the error term suggests that a spatial error model (SEM) is appropriate to address the residual spatial structure. This suggests that the model is failing to account for spatially-clustered latent drivers—such as regional tectonic patterns, spatially-dependent climatic factors not captured by the predictors, or even systematic biases in the inventory mapping. The higher significance of the LM-error test indicates that a model addressing spatially autocorrelated errors (e.g., SEM) may be more appropriate than one modeling endogenous interaction (e.g., SLy).

4.4 Spatial regression models

We evaluated landslide susceptibility regression models using two main approaches: (1) Spatial heterogeneity models, which allow regression coefficients to vary across space using spatial regime models (SR) and Geographically Weighted Regression (GWR) models, and (2) Spatial dependence models, which account for spatial interactions between neighboring units using spatial autoregressive models (SLy, SEM, SAC, SLX, SDEM, SDM, GNS). Finally, we combined both approaches, integrating spatial dependence and heterogeneity into a single framework (Table 2).

Table 2: Summary of the models used in the study and the figures/tables presenting their results.

Model	**Description**	**Results in**
(non-spatial)	Standard regression without spatial effects. Baseline model for comparison.	Table 1, Fig. 8
Spatial heterogeneity models		
Spatial Regime (SR)	Models with intercepts varying by regions (basins/clusters).	Fig. 9
	Models with intercepts and slopes varying by regions (basins/clusters).	Fig. 10

Model	**Description**	**Results in**
Geographically Weighted Regression (GWR)	Models with coefficients varying locally using several bandwidths (BW = 10, 50, 200, 500).	Fig. 11
	Models with coefficients varying locally using different bandwidths (Area BW=124, Elevation BW=302, Relief BW=504).	Fig. 12
Spatial dependence models		
SLy	Model with spatial autocorrelation in the dependent variable.	Table 3
SEM	Model with spatial autocorrelation in the errors.	Table 3, Fig. 14
SLX	Model with spatial autocorrelation in the predictors.	Table 3
SDEM	Spatial Durbin error model, combining SEM with spatial effects on predictors.	Table 3
SDM	Spatial Durbin model, combining Sly with spatial effects on predictors.	Table 3
SAC	Combined model with spatial dependence in both the dependent variable and error term.	Table 3, Table 4
GNS	Model combining spatial effects in the dependent variable, the errors and the predictors.	Table 3
Combined model	Integration of SR using clusters and autoregressive models (SAC, SEM, SDEM).	Fig. 13

4.4.1 Regression models for spatial heterogeneity

We implemented spatial fixed-effect (intercept-only) models independently for the basin-based regimes and the morphometric cluster-based regimes (Figure 9). This allowed us to separately evaluate how each spatial partitioning scheme accounts for regional variations in landslide occurrence. In the basin regime model, the effects for the Cauca and Magdalena basins have comparable magnitudes, suggesting similar regional influences on landslide abundance. In the cluster regime model, the spatial regimes for clusters B and D show stronger marginal effects than those of morphometric predictors such as area (A), mean elevation (E), and mean relief (H), indicating grouping catchments by these properties captures important spatial variability (Figure 10). For example, *H* has a much higher coefficient in the Magdalena and Cauca basins, while *E* is higher for the Atrato and Cauca basins. For the cluster regimes, *E* is not statistically significant in any cluster. However, the SR model based on cluster regimes has a slightly better performance. The SRs model for cluster regimes slightly raises adjusted R^2 and reduces AIC, with only a marginal difference to the best-performing OLS model.

A key result from the Geographically Weighted Regression (GWR) is a surface of local parameter estimates for each predictor (Figure 11). we observe a clear trade-off: smaller bandwidths (e.g., BW=10) yield a better statistical fit (higher Adj. R^2, lower AIC) but produce highly uncertain parameter estimates. Conversely, larger bandwidths (e.g., BW=500) provide more stable coefficients, though at the cost of a poorer model fit. These results indicate that lower bandwidths in GWR increase multicollinearity among predictor variables; as the estimation window shrinks, predictors often exhibit higher redundancy within the local subset than they do across the entire dataset. This leads to unstable parameter estimates, counterintuitive parameter signs, overconfident R^2 values, and inflated standard errors (Kyriazos & Poga, 2023).

Figure 12 shows the spatial variation in the coefficients of **A**, **E** and **H** for different bandwidths, and reveal spatial non-stationarity. We observe a positive relationship between **A** and landslide abundance throughout most of the region. The influence of **E** is stronger in the northeastern and southwestern parts of the study area, whereas the effect of **H** increases toward the northwest and decreases toward the southeast.

This observed spatial non-stationarity provides insights into the interplay of tectonic, climatic, and geomorphological processes that are typically masked by the averaged estimates of global models. The varying coefficient surfaces (Fig. 12) suggest that the drivers of landsliding are not uniform. For example, the increasing influence of H towards the northwest corresponds to the highly dissected, tectonically active terrain of the Cauca canyon (Cluster B). In contrast, the influence of E is strongest in the northeastern and southwestern high-altitude plateaus (Cluster D). A single, global coefficient for E or H (Table 1) is likely too simplistic. Landslide susceptibility in this region is controlled by different dominant processes in different geomorphological domains—a finding achievable through heterogeneity models like GWR or SR.

4.4.2 Regression models for spatial dependency

Table 3 summarizes the estimates of the spatial autoregressive (SAR) models. While the spatial autoregressive error models (SEM and SDEM) yield the lowest AIC values, the improvement in model fit compared to the other spatial models is modest. Despite accounting for spatial autocorrelation in these models, the overall difference among them is negligible.

The spatial lag of exogenous predictors refers to including neighboring values of the predictor variables, thus capturing potential spillover effects from surrounding areas. In theory, this allows the model to account for the influence that environmental factors in adjacent catchments may have on landslide abundance at a given location. For example, high rainfall accumulation in one catchment could affect slope saturation in neighboring catchments, indirectly contributing to landslide occurrence. We find that the coefficients for the spatially lagged predictors are statistically insignificant, indicating that the influence of predictors is primarily local rather than being influenced by those in neighboring areas.

Table 3: Model Results for SAR Models

Predictors	**SLy**	**SEM**	**SAC**	**SLX**	**SDEM**	**SDM**	**GNS**
Constant	**0.824** (0.08)	**1.819** (0.11)	**1.658** (0.23)	**1.841** (0.05)	**1.812** (0.11)	**0.687** (0.09)	**2.342** (0.31)
A	**0.530** (0.04)	**0.552** (0.04)	**0.558** (0.04)	**0.551** (0.05)	**0.561** (0.04)	**0.557** (0.04)	**0.560** (0.04)
E	**0.263** (0.05)	**0.539** (0.09)	**0.502** (0.10)	**0.440** (0.19)	**0.391** (0.15)	**0.409** (0.16)	**0.381** (0.15)
H	**0.281** (0.05)	**0.469** (0.08)	**0.457** (0.08)	**0.446** (0.12)	**0.395** (0.09)	**0.388** (0.10)	**0.410** (0.09)
Wx-A				0.047 (0.12)	0.020 (0.12)	**-0.317** (0.10)	0.185 (0.15)
Wx-E				0.076 (0.21)	0.167 (0.20)	-0.192 (0.18)	0.345 (0.23)
Wx-H				0.128 (0.14)	0.204 (0.14)	-0.157 (0.12)	**0.368** (0.18)
ρ	0.561		0.089			0.621	-0.295
λ		0.621	0.565		0.621		0.751
Nagelkerke R^2	0.62	0.628	0.628		0.631	0.631	0.633
AIC	1468.111	1456.317	1457.932	1615.105	1458.16	1458.759	1457.047

p<0.05, (standard errors)

Finally, we combined spatial dependency and spatial heterogeneity by integrating the spatial lag model (SAC), spatial error model (SEM), and Spatial Durbin Error Model (SDEM) with the spatial regime (SR) model using clusters as regimes (Figure 13). We selected the cluster-based regime for this combination because it outperformed the basin-based regime in other models (Figure 10), capturing more relevant morphometric variability. Yet, the combined model provides only marginal improvements over the standalone SAR models. The SAR models already explains much of the spatial structure in the data, limiting the additional benefit from explicitly modeling spatial heterogeneity through the cluster regimes. Although the cluster regime adds information about localized variations in morphometric conditions, the gains in predictive performance remain modest, indicating that spatial dependency explains more of the landslide occurrence in the study area.

Figure 14 shows the landslide abundance estimated by the SEM model and the distribution of residuals, which have no spatial structure (MI = -0.015 with p = 0.71).

Table 4 presents the global spillover effect for spatial autoregressive models with endogenous interaction effects of the response variable (SLy, SAC, SDM and GNS). The spillover effects must be interpreted with caution, avoiding a purely statistical reading. The indirect effects (spillovers) reported here do not imply that, for example, high relief in catchment *i* physically cause landslides in catchment *j*. Rather, these terms estimate the impact of spatially autocorrelated predictors. The positive indirect effects for E (0.152) and H (0.206) mean that catchments with high relief tend to be geographically clustered with other catchments that also have high relief (as supported by Moran's I for these predictors). The model is thus indicating that the landslide susceptibility in a given catchment is influenced by both its own attributes (direct effect) and those of its surrounding geomorphological neighborhood (indirect effect).

Table 4: Global spillover effect

Predictors	**Direct**	**Indirect**	**Total**
Mean Area (A)	0.561	0.070	0.631
Mean elevation (E)	0.419	0.152	0.572
mean relief (R)	0.401	0.206	0.608

5. Discussion

5.1 Limitations of the non-spatial framework

We used a standard linear regression as a baseline to examine the relationship between landslide abundance and several predictors. Although these models feature elevation, relief, and catchment area as key predictors, their overall explanatory power is limited (Table 1), likely due to the non-Gaussian distribution of the response variable.

One point for discussion is the apparent statistical insignificance of geology (G) and land use (Lc) in the parsimonious models, a finding that may seem physically counter-intuitive. We argue this result should not be interpreted as a lack of geomorphological relevance, but rather as a demonstration of scale-dependent effects in susceptibility modeling. At the slope-scale, geological and land-cover variables are often dominant. However, at the catchment-scale of this study (median area: 48 km²), their explanatory power appears to be overwhelmed by the dominant, macro-scale morphometric gradients (E and H). This scale-dependent effect is likely attributable to two factors. First, using the modal class for Lc and G within a large catchment unit neglects the variety of classes. Second, at the regional scale, processes related to tectonic uplift and climate, which are expressed through macro-scale topography (E and H), appear as first-order controls on landslide abundance. G and Lc may therefore represent second-order, local controls that are inadequately captured by this aggregated modeling approach, rather than irrelevant factors.

Finally, the OLS residuals show a clear, non-random spatial structure, with a significant global Moran's I of 0.376 ($p = 0.001$). This residual autocorrelation shows that the OLS model is misspecified, as it fails to capture local variations and violates the assumption of independent errors. This pattern is not uniform; exploratory analysis (Fig. 8) shows significant overestimation in the high-relief, complex terrain of clusters B and D .

5.2 Interpreting spatial structure

We interpret the two forms of spatial structure that our models addressed: heterogeneity and dependence. This is not merely a statistical exercise; it is a diagnostic effort to understand *why* the OLS model failed and what physical processes it obscured.

5.2.1 Interpreting spatial heterogeneity (GWR and SR models)

Model results confirm that the influence of geomorphological drivers is not uniform across the study area. By allowing coefficients to vary, GWR (Figs. 11, 12) and SR models (Figs. 9, 10) revealed distinct geomorphological domains where different processes are dominant.

For instance, the explanatory power of mean relief (H) is highest in the highly-incised, tectonically active terrain of the Cauca canyon (Cluster B), while mean Elevation (E) exerts a primary influence in the plateaus of the Central Cordillera (Cluster D). This demonstrates that a single, global coefficient for these predictors (as in Table 1) is a geomorphological oversimplification, averaging out the distinct processes that control landslide occurrence in different regions. Fotheringham et al. (2009) refer to this type of variability as different levels of spatial non-stationarity, where some predictors exhibit stronger local variations over short distances, while others may appear more stable.

While GWR offers a continuous surface of this variability, its results can be sensitive to bandwidth selection; narrow bandwidths risk parameter instability and multicollinearity, while wide bandwidths over-smooth local effects (Chen et al., 2024). While recognizing the potential for local instability in GWR estimates, these bandwidth variations suggest potential differences in the spatial scales at which each variable conditions landslide susceptibility. The relatively narrow bandwidth for variable **A** suggests that its influence might be more localized, potentially aligning with the spacing of catchment divides, whereas the moderate bandwidth for elevation may reflect broader regional topographic trends and their interaction with rainfall. In contrast, the larger bandwidth for mean relief (**H**) suggests a more consistent effect across broader areas, consistent with the smoother spatial variation of macro-scale morphometry compared to localized features. However, these scale-dependent interpretations remain exploratory, as the narrower bandwidths are also where the model is most sensitive to local collinearity. The SR model, while dependent on *a priori* region definition (e.g., our geomorphological clusters), avoids this instability and offers a clearer, more interpretable comparison of effects between these distinct process domains.

5.2.2 Interpreting spatial dependence (SAR models)

The spatial dependence models address the assumption of independent observations. Our analysis, guided by the LM tests (Section 4.3), found that the Spatial Error Model (SEM) had the most parsimonious fit (lowest AIC) among the tested SAR variants. A superior fit for a spatial lag model (e.g., SLy) would have implied a 'contagion' process, where high landslide frequency in one catchment directly influences the frequency in its neighbor. While a direct physical contagion for landslides—where one event triggers another across a catchment divide—is rare, this model can be appropriate if theory suggests a spatially contagious process is acting as a proxy for unobserved, clustered phenomena like regional deforestation patterns or cascading hillslope adjustments. The better performance of the SEM suggests this is not the dominant process, pointing instead to autocorrelated drivers as the most physically justifiable explanation.

The significant λ term in the SEM is effectively a proxy for latent, spatially-clustered drivers that were not included in our model. These could include unmeasured factors such as regional jointing patterns, localized storm cells not captured by the 5-km CHIRPS data, or systematic, spatially-clustered biases in the landslide inventory itself. Thus, the SEM not only corrects the model but also serves as a diagnostic tool to highlight the influence of these latent factors.

5.3 Practical implications for landslide susceptibility

The findings of this study have direct and practical implications for regional landslide susceptibility assessment, moving beyond statistical metrics to inform resource allocation and mitigation planning. First, our results challenge the validity of applying global susceptibility models to large, complex regions like the Andes. A non-spatial OLS model, for example, would incorrectly assign a single, uniform weight to relief for the entire study area. A regional susceptibility manager using this global model might misallocate resources by focusing on high-relief areas in regions where, as our GWR and SR results demonstrate (Figs. 10, 12), Mean Elevation (E) is the true dominant factor (e.g., Cluster D). Second, the strong evidence for spatial heterogeneity (non-stationarity) provided by the GWR and SR models suggests that a global mitigation strategy is inherently flawed. Instead, susceptibility mapping and subsequent land-use planning should be stratified by the distinct geomorphological domains identified in our clustering analysis (Fig. 7). Mitigation efforts in the highly-incised Cauca canyon (Cluster B) should prioritize factors related to high local relief and associated slope processes, while efforts in the Central Cordillera (Cluster D) may need to focus on processes occurring at high elevations, which may be linked to different rainfall patterns or bedrock properties.

Finally, the better performance of the Spatial Error Model (SEM) serves as a diagnostic tool for practitioners. It provides a quantitative warning that the selected, measured predictors (A, E, H), while significant, are insufficient to fully explain the observed landslide patterns. This finding highlights the need to invest in acquiring better, more granular data on the latent drivers the SEM is capturing. This could include, for example, high-resolution rainfall data (to capture localized storms), detailed lithological or structural maps (to capture regional joint patterns), or improved inventory mapping protocols to reduce systematic spatial bias.

5.4 Methodological considerations and future research

While our findings demonstrate the value of integrating spatial structure, several methodological considerations and avenues for future research must be addressed.

(i) The quality and potential biases of the landslide inventory are a primary concern. Any spatial model will amplify, not correct, systematic spatial biases present in the data, such as a concentration of mapped landslides along infrastructure (Caleca et al., 2025). While spatial models can be strategically used to isolate this bias, our study proceeds with the assumption that the inventory, while incomplete, is a spatially representative sample. Future work should explicitly test the sensitivity of these models to known inventory biases.

(ii) The definition of the spatial weights matrix (W) is a subjective step. Our data-driven choice of a k=5 nearest-neighbor matrix was based on maximizing spatial autocorrelation. However, this choice is not neutral; it pre-defines the scale and nature of spatial interactions. Future research could explore physically-grounded matrices based on, for example, hydrological connectivity or geomorphological similarity, to test specific process-based hypotheses.

(iii) Our justification for selecting the SEM is further supported by a robust statistical model selection process. When we tested combined models (e.g., SAC) that include terms for both error (λ) and endogenous (ρ) dependence, we found that the spatial lag coefficient (ρ) decreased significantly while the spatial error coefficient (λ) remained stable and significant. This strongly indicates that spatial dependence in our data is primarily captured by the error term, reinforcing our conclusion that the process is driven by latent, unmeasured spatial factors rather than a true contagion effect.

(iv) While we explored models for heterogeneity (SR) and dependence (SEM) separately, we also tested combined frameworks (e.g., SR-SEM)8. This approach, which separates large-scale heterogeneity from fine-scale dependence, introduces significant complexity, multicollinearity, and overfitting. In our case, the gains in performance were marginal. However, future work should explore hybrid methods that can more adaptively model these dual spatial effects, especially in highly complex terrains.

(v) This study was conducted within a Gaussian-based regression framework, requiring a `log(Y+1)` transformation of the discrete, zero-inflated landslide count data. We acknowledge that this transformation is an approximation and does not fully account for the underlying data-generating process. This approach was adopted due to the practical availability of robust spatial econometric software. A critical avenue for future research is the application of more sophisticated spatial regression frameworks explicitly designed for count data, such as Spatial Poisson, Negative Binomial (NB), or Zero-Inflated (ZINB) models. Such models would provide a more statistically rigorous estimation of landslide frequency, eliminating the need for data transformation and offering deeper insights into the complex spatial patterns of landslide susceptiblity.

(vi) Finally, while our analysis identified the SEM as the most parsimonious model, other variants like the Spatial Durbin Model (SDM) or Durbin Error Model (SDEM) remain highly relevant for future research. These models, which account for exogenous spillovers, are ideal when observation suggests strong spatial interactions between predictors. For example, they could explicitly model processes such as hydrological connectivity (where high rainfall in an upstream catchment affects a downstream one) or infrastructure interdependence (where a road network propagates risk). Exploring these models could offer even deeper insights where such cross-catchment processes are dominant.

6. Conclusion

Our analysis demonstrates that the spatial structure of landslide data, often ignored in susceptibility modeling, is not a statistical nuisance but a fundamental component of the geomorphological system. We have shown that standard, non-spatial models are fundamentally misspecified, as evidenced by significant residual autocorrelation and their failure to capture scale-dependent effects.

By integrating spatial models, we move beyond a single, global model to reveal that landslide controls are spatially heterogeneous, with relief dominating in high-incision terrains and elevation controlling susceptibility on high-altitude plateaus. Furthermore, the dominance of spatial error (SEM) provides a powerful diagnostic, suggesting that landslide patterns are strongly driven by latent, spatially-clustered factors, such as tectonic fabrics, localized storm cells, or inventory biases

We conclude that explicitly accounting for both spatial heterogeneity and dependence is a necessary step to advance landslide susceptibility from a statistical pattern-matching exercise to a more realistic, process-based susceptibility assessment. Future work must build on this by integrating non-Gaussian frameworks and by using spatial models to strategically investigate the impact of inventory quality on susceptibility assessment.

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Competing Interests

The authors declare no competing interests.

Author Contributions

EA designed the study, compiled the landslide inventory, and led the spatial modeling and statistical analysis. LL and OK contributed to the conceptual framework of spatial models and supervised the implementation of spatial regression techniques. OK revised the interpretation of results, and supported the writing and structuring of the discussion. All authors contributed to the manuscript preparation, reviewed, and approved the final version.

Open Research

Data and code used for this study are fully available in Github (https://github.com/edieraristizabal/PAPER_SAR).

References

Alvioli, M., Marchesini, I., Reichenbach, P., Rossi, M., Ardizzone, F., Fiorucci, F., & Guzzetti, F. (2016). Automatic delineation of geomorphological slope units with r. slopeunits v1. 0 and their optimization for landslide susceptibility modeling. *Geoscientific Model Development*, *9*(11), 3975–3991.

Anselin, L. (1988). *Spatial econometrics: methods and models*. Kluwer Academic Publishers.

Anselin, L. (1990). Spatial dependence and spatial structural instability in applied regression analysis. *Journal of Regional science*, *30*(2), 185–207.

Anselin, L. (1996). Simple diagnostic tests for spatial dependence. *Regional science and urban economics*, *26*(1), 77–104.

Anselin, L. (2022). Spatial econometrics. In *Handbook of spatial analysis in the social sciences* (pp. 101–122). Edward Elgar Publishing.

Arbia, G. (2016). Spatial econometrics: A rapidly evolving discipline. *Econometrics*, *4*(1), 18.

Aristizábal, E., & Korup, O. (2025). Linking landslide patterns to transient landscapes in the northern Colombian Andes. *JGR: Earth sciences*, xxx.

Aristizábal, E., & Sánchez, O. (2020). Spatial and temporal patterns and the socioeconomic impacts of landslides in the tropical and mountainous Colombian Andes. *Disasters, 44*(3), 596–618.

Ba, Q., Chen, Y., Deng, S., Yang, J., & Li, H. (2018). A comparison of slope units and grid cells as mapping units for landslide susceptibility assessment. *Earth Science Informatics, 11*, 373–388.

Bai, S., Lu, P., & Wang, J. (2015). Landslide susceptibility assessment of the Youfang catchment using logistic regression. *Journal of Mountain Science, 12*, 816–827.

Banerjee, S., Gelfand, A. E., Finley, A. O., & Sang, H. (2008). Gaussian predictive process models for large spatial data sets. *Journal of the Royal Statistical Society Series B: Statistical Methodology, 70*(4), 825–848.

Bivand, R. (2022). R Packages for Analyzing Spatial Data: A Comparative Case Study with Areal Data. *Geographical Analysis, 54*(3), 488-518.

Bivand, R., Millo, G., & Piras, G. (2021). A Review of Software for Spatial Econometrics in R. *Mathematics, 9*(11), 1276.

Bonham-Carter, G. F., Agterberg, F. P., & Wright, D. F. (1989). Integration of geological datasets for gold exploration in Nova Scotia. *Digital geologic and geographic information systems*, *10*, 15–23.

Brabb, E. E. (1984). Innovative approaches to landslide hazard and risk mapping.

Breiman, L. (2001). *Random Forests*. Springer.

Brunsdon, C., Fotheringham, A. S., & Charlton, M. E. (1996). Geographically weighted regression: a method for exploring spatial nonstationarity. *Geographical analysis*, *28*(4), 281–298.

Bryce, E., Castro-Camilo, D., Dashwood, C., Tanyas, H., Ciurean, R., Novellino, A., & Lombardo, L. (2024). An updated landslide susceptibility model and a log-Gaussian Cox process extension for Scotland. *Landslides*, 1–19.

Bryce, E., Lombardo, L., Van Westen, C., Tanyas, H., & Castro-Camilo, D. (2022). Unified landslide hazard assessment using hurdle models: a case study in the Island of Dominica. *Stochastic Environmental Research and Risk Assessment*, *36*(8), 2071–2084.

Caleca, F., Lombardo, L., Steger, S., Tanyas, H., Raspini, F., Dahal, A., ... & others. (2025). Pan-European landslide risk assessment: From theory to practice. *Reviews of Geophysics*, *63*(1), e2023RG000825.

Chen, F., Leung, Y., Wang, Q., & Zhou, Y. (2024). Spatial non-stationarity test of regression relationships in the multiscale geographically weighted regression model. *Spatial Statistics*, 100846.

Cliff, A. D., & Ord, J. K. (1981). *Spatial processes: models & applications*.

Corominas, J., van Westen, C., Frattini, P., Cascini, L., Malet, J.-P., Fotopoulou, S., ... & others. (2014). Recommendations for the quantitative analysis of landslide risk. *Bulletin of engineering geology and the environment*, *73*, 209–263.

Cressie, N. (2015). *Statistics for spatial data*. John Wiley & Sons.

Di Napoli, M., Tanyas, H., Castro-Camilo, D., Calcaterra, D., Cevasco, A., Di Martire, D., ... & Lombardo, L. (2023). On the estimation of landslide intensity, hazard and density via data-driven models. *Natural Hazards, 119*(3), 1513–1530.

Durmaz, M., Hürlimann, M., Huvaj, N., & Medina, V. (2023). Comparison of different hydrological and stability assumptions for physically-based modeling of shallow landslides. *Engineering Geology, 323*, 107237.

Elhorst, J. P. (2013). On spatial econometric models, spillover effects, and W. *Louvain-la-Neuve: European Regional Science Association (ERSA)*.

Elhorst, J. P. (2022). The dynamic general nesting spatial econometric model for spatial panels with common factors: Further raising the bar. *Review of Regional Research, 42*(3), 249–267.

Fang, Z., Wang, J., Wang, Y., Du, B., & Liu, G. (2024). Improved landslide prediction by considering continuous and discrete spatial dependency. *Landslides*, 1–16.

Fell, R., Corominas, J., Bonnard, C., Cascini, L., Leroi, E., & Savage, W. Z. (2008). Guidelines for landslide susceptibility, hazard and risk zoning for land use planning. *Engineering geology, 102*(3-4), 85–98.

Fischer, M. M., & Wang, J. (2011). *Spatial data analysis: models, methods and techniques*. Springer Science & Business Media.

Fletcher, R., & Fortin, M.-J. (2018). Spatial dependence and autocorrelation. *Spatial ecology and conservation modeling: applications with R*, 133–168.

Fotheringham, A. S., Brunsdon, C., & Charlton, M. (2000). *Quantitative geography: perspectives on spatial data analysis*. Sage.

Fotheringham, A. S., Brunsdon, C., & Charlton, M. E. (2009). Geographically weighted regression. *The Sage handbook of spatial analysis, 1*, 243–254.

Funk, C., Peterson, P., & Landsfeld, M. (2015). The Climate Hazard Group InfraRed Precipitation with Station Data (CHIRPS). *Bulletin of the American Meteorological Society*, *96*, 1401-1419.

Getis, A. (2009). Spatial weights matrices. *Geographical Analysis, 41*(4), 404–410.

Getis, A., & Aldstadt, J. (2004). Constructing the spatial weights matrix using a local statistic. *Geographical analysis, 36*(2), 90–104.

Haining, R. P., & Li, G. (2020). *Regression Modelling Wih Spatial and Spatial-Temporal Data: A Bayesian Approach*. CRC Press.

Jaya, I., & Chadidjah, A. (2021). Spatial Autoregressive in Ecological Studies: A Comparison of the SAR and CAR Models. *Engineering Letters, 29*(1).

Kaiser, H. F. (1960). The application of electronic computers to factor analysis. *Educational and Psychological Measurement, 20*(1), 141–151.

Kelejian, H. H., & Prucha, I. R. (2007). The relative efficiencies of various predictors in spatial econometric models containing spatial lags. *Regional Science and Urban Economics, 37*(3), 363–374.

Kriegler, F. J. (1969). Normalized Difference Vegetation Index (NDVI). *Journal of Forestry Research*, *5*, 112-116.

Kyriazos, T., & Poga, M. (2023). Dealing with multicollinearity in factor analysis: the problem, detections, and solutions. *Open Journal of Statistics, 13*(3), 404–424.

LeSage, J. P. (2014). Spatial econometric panel data model specification: A Bayesian approach. *Spatial statistics, 9*, 122–145.

LeSage, J., & Pace, R. K. (2009). *Introduction to spatial econometrics*. Chapman and Hall/CRC.

Li, K., Liu, J., Xue, Y., Rahman, S., & Sriboonchitta, S. (2022). Consequences of ignoring dependent error components and heterogeneity in a stochastic frontier model: an application to rice producers in northern Thailand. *Agriculture, 12*(8), 1078.

Li, P., Shen, Q., & Guo, Y. (2014). Bare Soil Index for Land Cover Analysis. *Journal of Applied Remote Sensing, 8*, 123-134.

Liu, Q., Zhao, Q., Lan, Q., Huang, C., Yang, X., & Tang, Z. (2024). Regional dynamic hazard assessment of rainfall–induced landslide guided by geographic similarity. *Bulletin of Engineering Geology and the Environment*, *83*(12), 1–22.

Logan, T. L., & Karr, C. L. (2014). Advanced Land Observing Satellite (ALOS) - Phased Array-Type L-Band Synthetic Aperture Radar (PALSAR). *Journal of Geospatial Research, 7*, 102-119.

Lombardo, L., Opitz, T., Ardizzone, F., Guzzetti, F., & Huser, R. (2020). Space-time landslide predictive modelling. *Earth-science reviews*, *209*, 103318.

Nagelkerke, N. J. D. (1991). A note on a general definition of the coefficient of determination. *Biometrika, 78*(3), 691–692.

Opitz, T., Bakka, H., Huser, R., & Lombardo, L. (2022). High-resolution Bayesian mapping of landslide hazard with unobserved trigger event. *The Annals of Applied Statistics, 16*(3), 1653–1675.

Oshan, T. M., Li, Z., Kang, W., Wolf, L. J., & Fotheringham, A. S. (2019). mgwr: A Python implementation of multiscale geographically weighted regression for investigating process spatial heterogeneity and scale. *ISPRS International Journal of Geo-Information, 8*(6), 269.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research, 12,* 2825-2830.

Peng, Z., & Inoue, R. (2024). Multiscale Continuous and Discrete Spatial Heterogeneity Analysis: The Development of a Local Model Combining Eigenvector Spatial Filters and Generalized Lasso Penalties. *Geographical Analysis, 56*(2), 303–327.

Piras, G., & Sarrias, M. (2023). Heterogeneous spatial models in R: spatial regimes models. *Journal of Spatial Econometrics, 4*(1), 4.

Quevedo, R. P., Maciel, D. A., Uehara, T. D. T., Vojtek, M., Renno, C. D., Pradhan, B., ... & Pham, Q. B. (2022). Consideration of spatial heterogeneity in landslide susceptibility mapping using geographical random forest model. *Geocarto International, 37*(25), 8190–8213.

Reichenbach, P., Rossi, M., Malamud, B. D., Mihir, M., & Guzzetti, F. (2018). A review of statistically-based landslide susceptibility models. *Earth-science reviews, 180*, 60–91.

Rey, S. J., & Anselin, L. (2007). PySAL: A Python Library of Spatial Analytical Methods. *The Review of Regional Studies, 37*(1), 5-27.

Rey, S., Arribas-Bel, D., & Wolf, L. J. (2023). *Geographic data science with python*. Chapman and Hall/CRC.

Ripley, B. D. (1988). *Statistical inference for spatial processes*. Cambridge university press.

Rousseeuw, P. J. (1987). Silhouettes: A Graphical Aid to the Interpretation and Validation of Cluster Analysis. *Journal of Computational and Applied Mathematics, 20*, 53-65.

Rue, H., & Held, L. (2005). *Gaussian Markov random fields: theory and applications*. Chapman and Hall/CRC.

Schabenberger, O., & Pierce, F. J. (2001). *Contemporary statistical models for the plant and soil sciences*. CRC press.

Smith, H. G., Neverman, A. J., Betts, H., & Spiekermann, R. (2023). The influence of spatial patterns in rainfall on shallow landslides. Geomorphology, 437, 108795.

Stakhovych, S., & Bijmolt, T. H. (2009). Specification of spatial models: A simulation study on weights matrices. *Papers in Regional Science*, *88*(2), 389–409.

Thorndike, R. L. (1953). Who Belongs in the Family? *Psychometrika*, *18*, 267-276.

Van Westen, C. J., Castellanos, E., & Kuriakose, S. L. (2008). Spatial data for landslide susceptibility, hazard, and vulnerability assessment: An overview. *Engineering geology*, *102*(3-4), 112–131.

Ver Hoef, J. M., Peterson, E. E., Hooten, M. B., Hanks, E. M., & Fortin, M.-J. (2018). Spatial autoregressive models for statistical inference from ecological data. *Ecological Monographs*, *88*(1), 36–59.

Villalta, P., & Smith, R. (2002). LINE Algorithms for Hillshade Extraction Using SRTM Data. *International Journal of Remote Sensing*, *23*, 457-470.

Wall, M. M. (2004). A close look at the spatial structure implied by the CAR and SAR models. *Journal of statistical planning and inference*, *121*(2), 311–324.

Wang, J., Haining, R., Zhang, T., Xu, C., & Hu, M. (2022). *Statistics for Spatially Stratified Heterogeneous Data*. arXiv preprint arXiv:2211.16918.

Wickham, H. (2008). *Applied spatial data analysis with r*. Springer.

Xiao, T., Yu, L., Tian, W., Zhou, C., & Wang, L. (2021). Reducing local correlations among causal factor classifications as a strategy to improve landslide susceptibility mapping. *Frontiers in Earth Science*, *9*, 781674.

Xu, H. (2006). Modification of Normalized Difference Water Index (MNDWI) to Enhance Open Water Features in Remotely Sensed Imagery. *International Journal of Remote Sensing*, *27*, 3025-3033.

Yadav, R., Huser, R., Opitz, T., & Lombardo, L. (2023). Joint modelling of landslide counts and sizes using spatial marked point processes with sub-asymptotic mark distributions. *Journal of the Royal Statistical Society Series C: Applied Statistics*, *72*(5), 1139–1161.

Zha, Y., Gao, J., & Ni, S. (2003). Use of Normalized Difference Built-up Index in Automated Urban Area Mapping. *Remote Sensing of Environment*, *87*, 467-480.

Zhang, T., Cheng, C., & Wu, X. (2023). Mapping the spatial heterogeneity of global land use and land cover from 2020 to 2100 at a 1 km resolution. *Scientific Data*, *10*(1), 748.

Fig. 1 Overview of spatial regression models that incorporate spatial heterogeneity (pink) and spatial dependency (blue). Spatially-lagged Conditional Autoregressive Models (CAR) or Simultaneous Autoregressive models (SAR) target specific spatial structures: CAR models include options like Intrinsic CAR, Besag-York-Mollie (BYM), and Leroux, while SAR models account for spatial lag in spatial connectivity or network structures; by adding terms for spatial autocorrelation in the response variable WX, the local error $\lambda W\mu$, or spatial lag effects ρWy, we obtain different variants of spatial regression models.

Fig. 2 Location of the study area and 13,777 landslides between 1970 and 2023. The first map (A) illustrates the spatial distribution of individual landslides across the study region. The inset shows the geographical location of the study area within South America. Map B shows the abundance of landslides aggregated by catchments.

Fig. 3 Histogram of landslide abundance per catchment (A), highlighting high skewness with most observations clustered near zero, i.e. no detectable landslides in a given catchment. The panel B shows transformed variable, i.e. log(Landslide abundance + 1); black line is a Gaussian kernel density estimate.

Fig. 4 Principal Component Analysis (PCA) biplot of catchment characteristics (the hypsometric integral (Hi), mean elevation (E), mean slope S, mean local relief (H), lineament density ((Ld), mean annual rainfall (P), land use cover (Lc), geology (G), the number of days with accumulated daily rainfall exceeding 20 mm (P20). The first two principal components (PC1 and PC2), explain 49% and 21% of the total variance. Each bubble represents a catchment, with red circles indicating catchments with reported landslides (Landslides > 0) and green circles indicating those without (Landslides = 0). Bubble size is scaled to the number of landslides in each catchment; arrows show the loading of catchment variables on the principal components.

Fig. 5 Five–nearest-neighbour (k = 5) connectivity representing the neighboring relationships between catchments in the study area (blue nodes = centroids, red edges = neighbour links). Spatial connectivity network.

Fig. 6 Local Indicator of Spatial Autocorrelation (LISA) and the Moran Index for the slope angle. Red color means high values surrounded by high values (Q1-hotspot), blue color means low values surrounded by low values (Q3-coldspot), blue and yellow color mean high values-low values (Q2) and low values-high values (Q4), respectively. Gray dots indicate locations (catchments) where the spatial autocorrelation is statistically insignificant.

Fig. 7 Spatial regimes used in the study area: (A) basin boundaries and (B) morphometric clusters of 533 catchments. The left map shows the delineated boundaries of the major river basins in the study area, including the Atrato, Cauca, and Magdalena basins, represented by green, red, and blue, respectively. These basins constitute the first spatial regime. The right map displays the second spatial regime, derived from a cluster analysis of morphometric properties using a K-Nearest Neighbors (KNN) algorithm (K = 5), which identified four clusters (A, B, C, and D) that group catchments with similar morphometric characteristics.

Fig. 8 Non-spatial OLS model standardized residuals for basins and clusters. The top panels (A, B) show the spatial distribution of model residuals across the study area. Positive residuals (in blue) indicate overestimation by the model, while negative residuals (in red) indicate underestimation. The polygons' outline in the map is color coded according to the three basins. The panels C and D show box plots of standardized residuals across the Atrato, Cauca, and Magdalena basins: non-overlapping notches identify differences in median values.

Fig. 9 Spatial fixed effects with estimated standard errors for the cluster and basin-regime models. The panel A displays results for the cluster regimes, and the panel B shows the basin regimes.

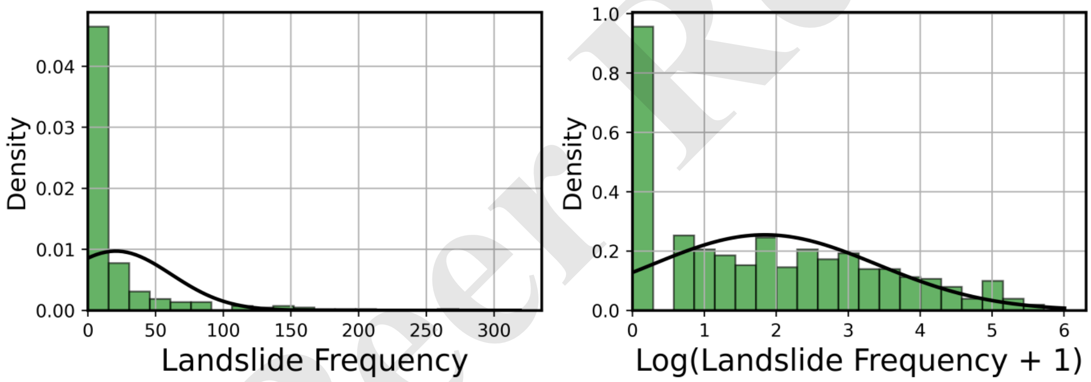
Fig. 10 Group-level intercepts and slope models for spatial regimes (Basin and Cluster), showing parameter estimates with standard errors.

Fig. 11 Parameter estimates for a Geographically Weighted Regression (intercept, Area (A), Elevation (E), Relief (H) with varying bandwidths in terms of number of catchments (BW = 10, BW = 50, BW = 200, BW = 500). Points are the means, and bars indicate the standard error for each predictor within each bandwidth.

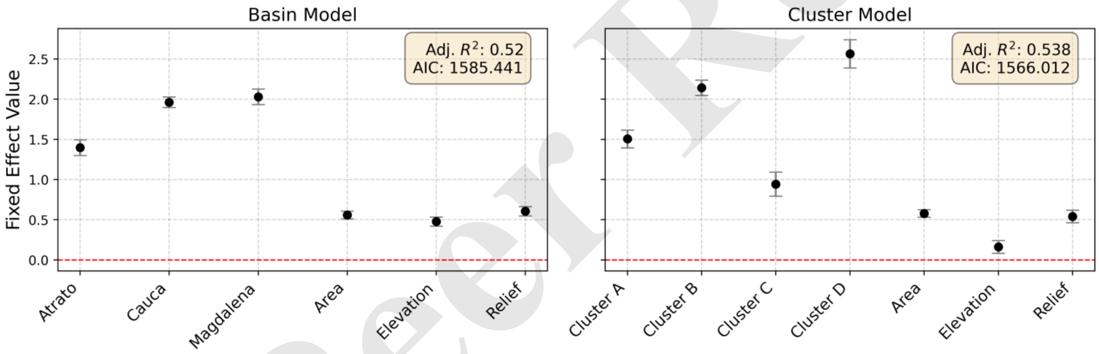
Fig. 12 Spatial variation of Multiscale Geographically Weighted Regression coefficients for (A) catchment area (BW = 124), (B) mean elevation (BW = 302), and (C) mean relief (BW = 504). BW in terms of neighbor catchments for each predictor were iteratively optimized in an adaptive process.

Fig. 13 Parameter estimates for SEM, SLy, and SDM models. Bubbles are mean estimates, bars are their standard errors. Other model-specific parameters are: λ for SEM (blue); ρ for SLy (green); and SDM (red), along with Nagelkerke R2 values for SLy and SDM models, and AIC values for each model. WX corresponds to the spatially lagged predictors, which captures the exogenous interaction effects among the predictors.

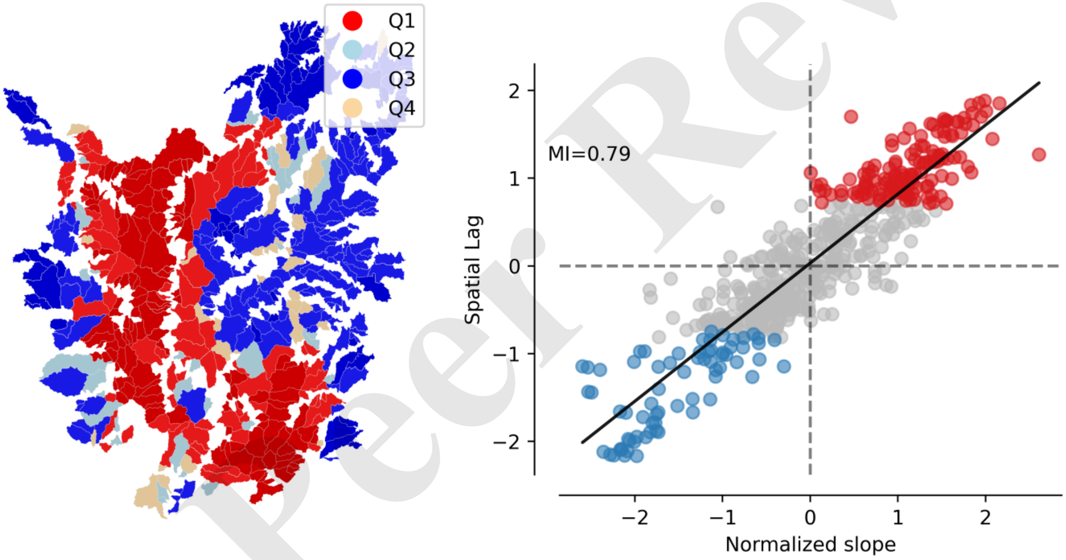
Fig. 14 Maps (A) of SEM residuals color-coded to the deviations between predicted and observed landslide occurrences. Negative residuals (blue) show underprediction, while positive residuals (yellow) show overprediction. Map (B) of the expected landslide abundance.

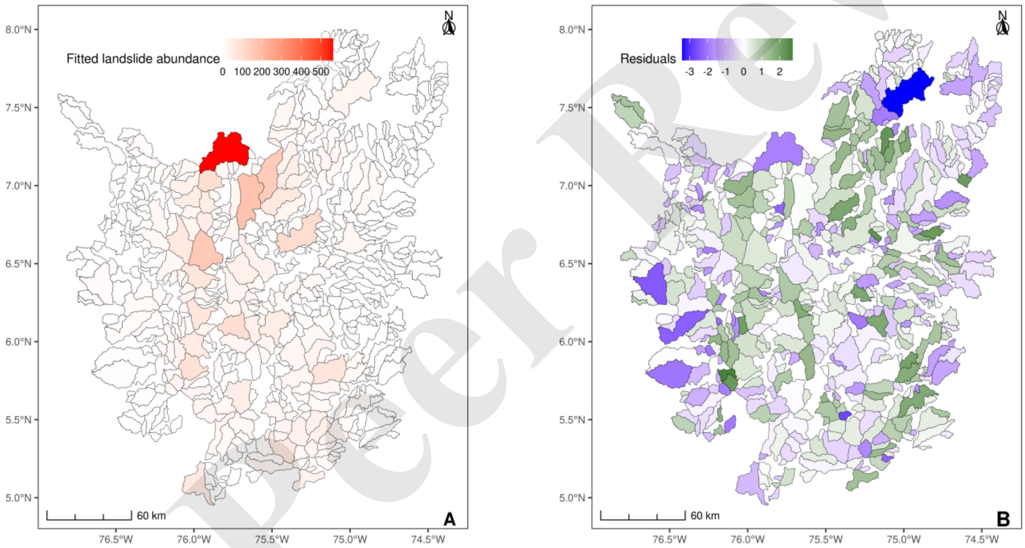


Density
0.04
0.03
0.02
0.01
0.00
0
50
100
150
200
250
300
Landslide Frequency
1.0
0.8
0.6
0.4
0.2
0.0
1
2
3
4
5
6
Log(Landslide Frequency + 1)

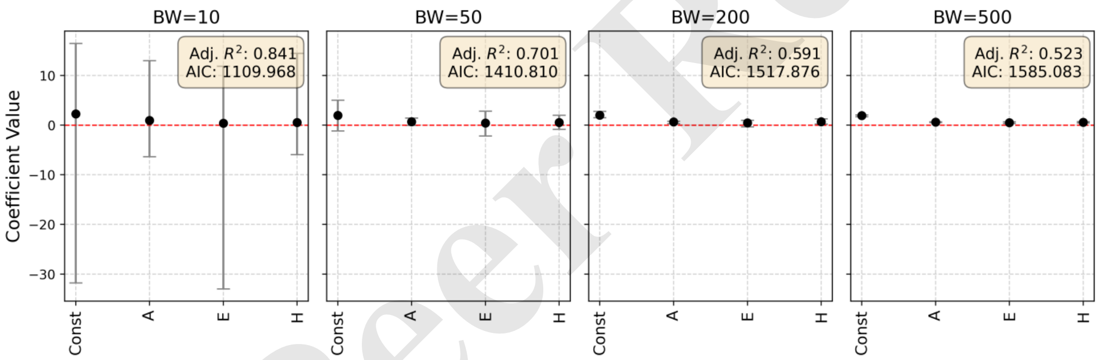


Basin Model
Cluster Model
Fixed Effect Value
Adj. R^2: 0.52
AIC: 1585.441
Adj. R^2: 0.538
AIC: 1566.012
Atrato
Cauca
Magdalena
Area
Elevation
Relief
Cluster A
Cluster B
Cluster C
Cluster D
Area
Elevation
Relief

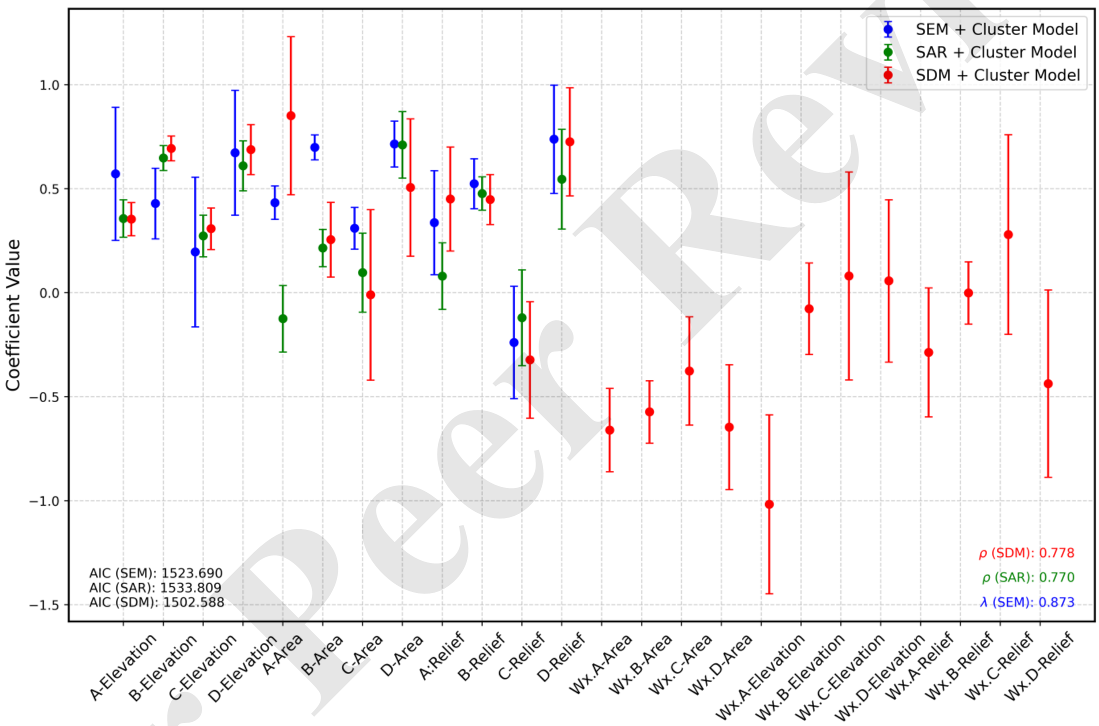

Q1
Q2
Q3
Q4
MI=0.79
Spatial Lag
Normalized slope
2
1
0
−1
−2
−2
−1
0
1
2

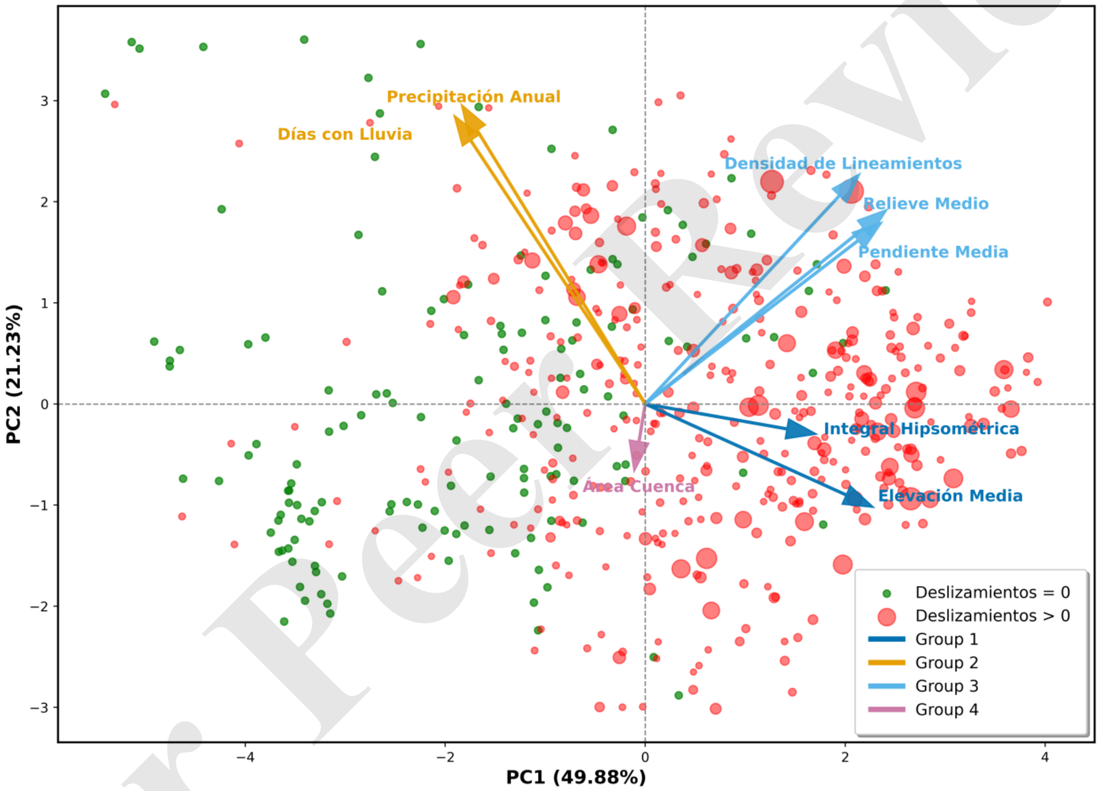


Fitted landslide abundance
0 100 200 300 400 500
Residuals
-3 -2 -1 0 1 2
8.0°N
7.5°N
7.0°N
6.5°N
6.0°N
5.5°N
5.0°N
76.5°W 76.0°W 75.5°W 75.0°W 74.5°W
60 km
N
A
B

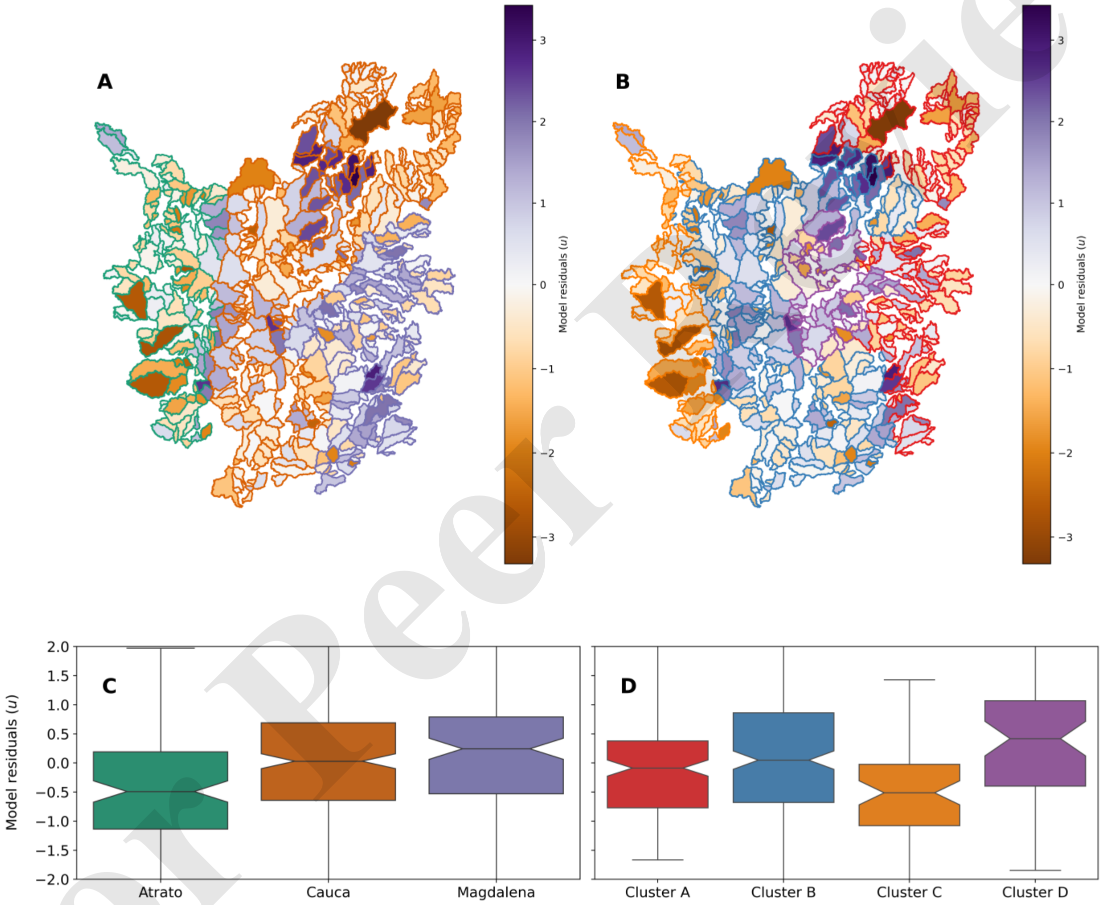


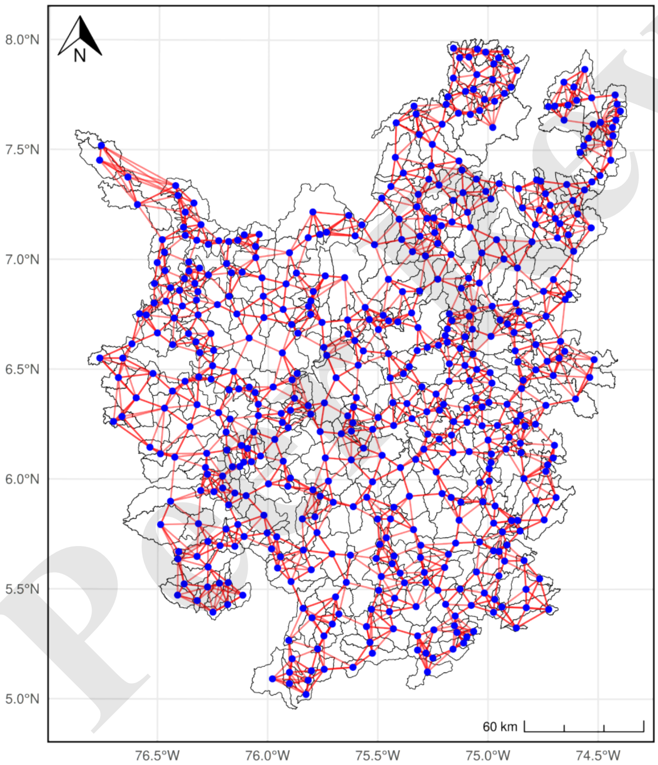
BW=10
BW=50
BW=200
BW=500
Adj. R^2: 0.841
AIC: 1109.968
Adj. R^2: 0.701
AIC: 1410.810
Adj. R^2: 0.591
AIC: 1517.876
Adj. R^2: 0.523
AIC: 1585.083
Coefficient Value
10
0
−10
−20
−30
Const
A
E
H

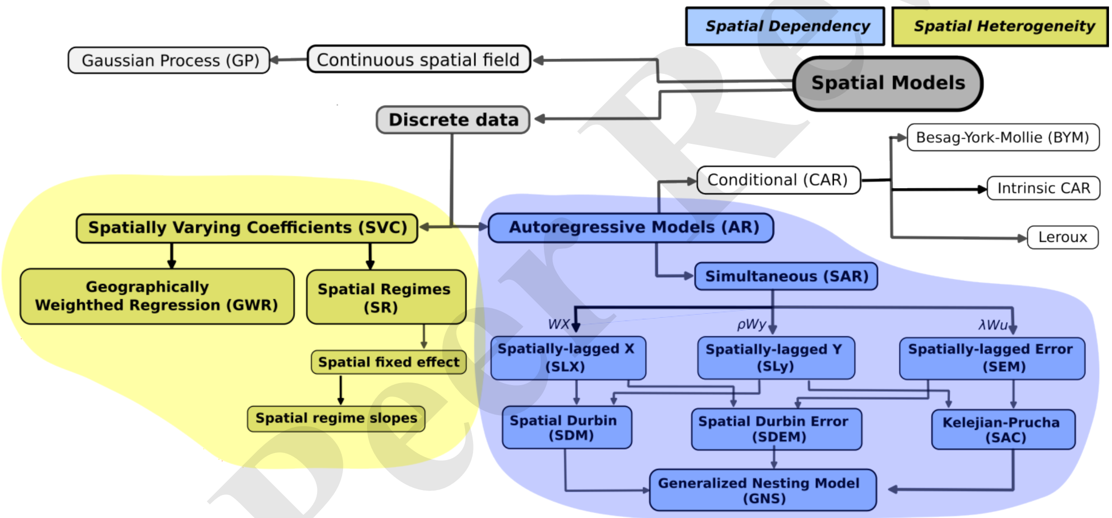

SEM + Cluster Model
SAR + Cluster Model
SDM + Cluster Model
Coefficient Value
1.0
0.5
0.0
−0.5
−1.0
−1.5
AIC (SEM): 1523.690
AIC (SAR): 1533.809
AIC (SDM): 1502.588
ρ (SDM): 0.778
ρ (SAR): 0.770
λ (SEM): 0.873
A-Elevation
B-Elevation
C-Elevation
D-Elevation
A-Area
B-Area
C-Area
D-Area
A:Relief
B-Relief
C-Relief
D-Relief
Wx.A-Area
Wx.B-Area
Wx.C-Area
Wx.D-Area
Wx.A-Elevation
Wx.B-Elevation
Wx.C-Elevation
Wx.D-Elevation
Wx.A-Relief
Wx.B-Relief
Wx.C-Relief
Wx.D-Relief

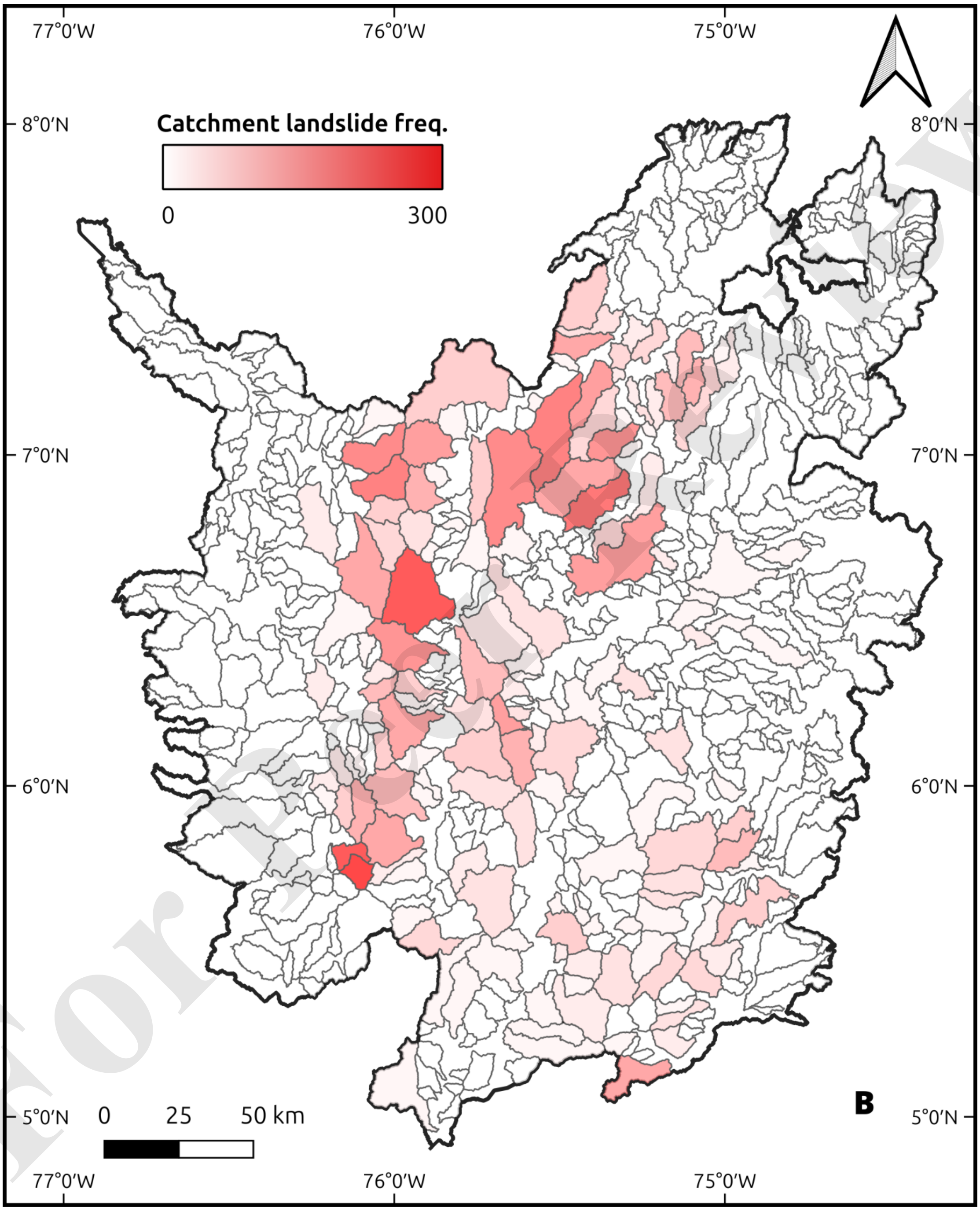


Precipitación Anual
Días con Lluvia
Densidad de Lineamientos
Relieve Medio
Pendiente Media
Integral Hipsométrica
Área Cuenca
Elevación Media
Deslizamientos = 0
Deslizamientos > 0
Group 1
Group 2
Group 3
Group 4
PC2 (21.23%)
PC1 (49.88%)

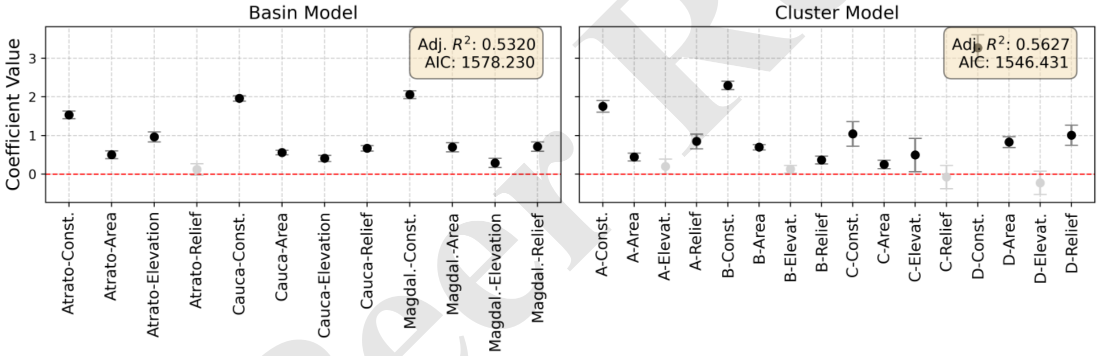

A
B
Model residuals (u)
3
2
1
0
−1
−2
−3
C
D
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
Atrato
Cauca
Magdalena
Cluster A
Cluster B
Cluster C
Cluster D


N
8.0°N
7.5°N
7.0°N
6.5°N
6.0°N
5.5°N
5.0°N
76.5°W
76.0°W
75.5°W
75.0°W
74.5°W
60 km

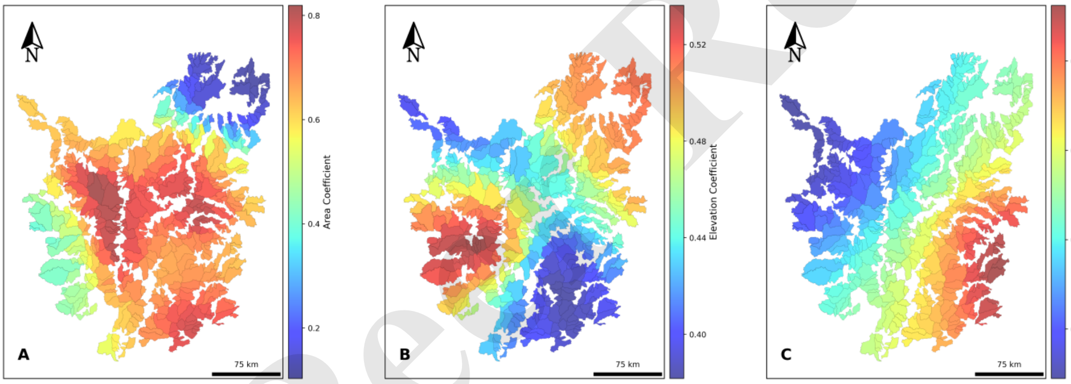

Spatial Dependency
Spatial Heterogeneity
Spatial Models
Gaussian Process (GP)
Continuous spatial field
Discrete data
Besag-York-Mollie (BYM)
Conditional (CAR)
Intrinsic CAR
Leroux
Spatially Varying Coefficients (SVC)
Autoregressive Models (AR)
Geographically Weighthed Regression (GWR)
Spatial Regimes (SR)
Simultaneous (SAR)
Spatial fixed effect
Spatial regime slopes
WX
ρWy
λWu
Spatially-lagged X (SLX)
Spatially-lagged Y (SLy)
Spatially-lagged Error (SEM)
Spatial Durbin (SDM)
Spatial Durbin Error (SDEM)
Kelejian-Prucha (SAC)
Generalized Nesting Model (GNS)

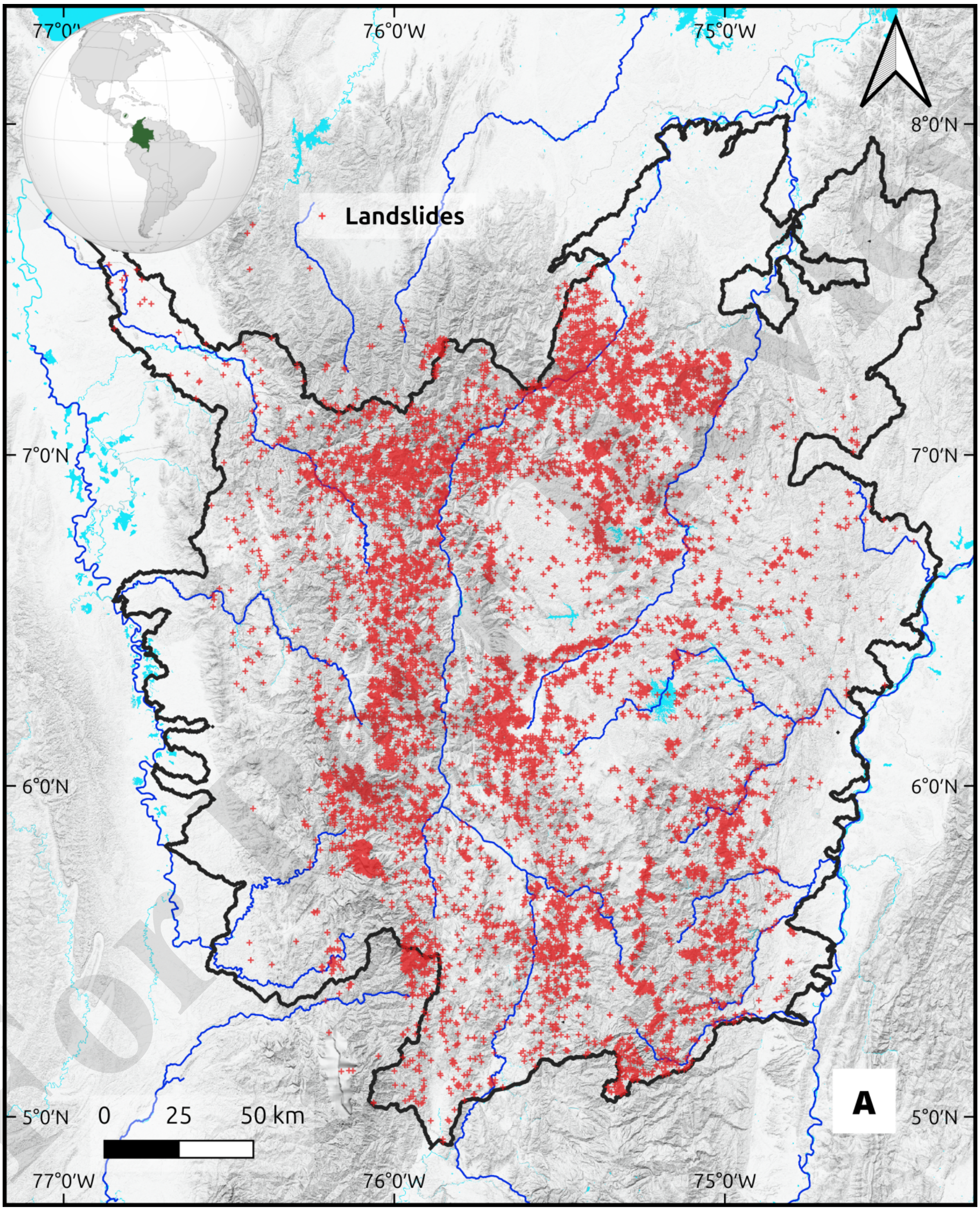


77°0'W
76°0'W
75°0'W
8°0'N
7°0'N
6°0'N
5°0'N
Catchment landslide freq.
0
300
0
25
50 km
B

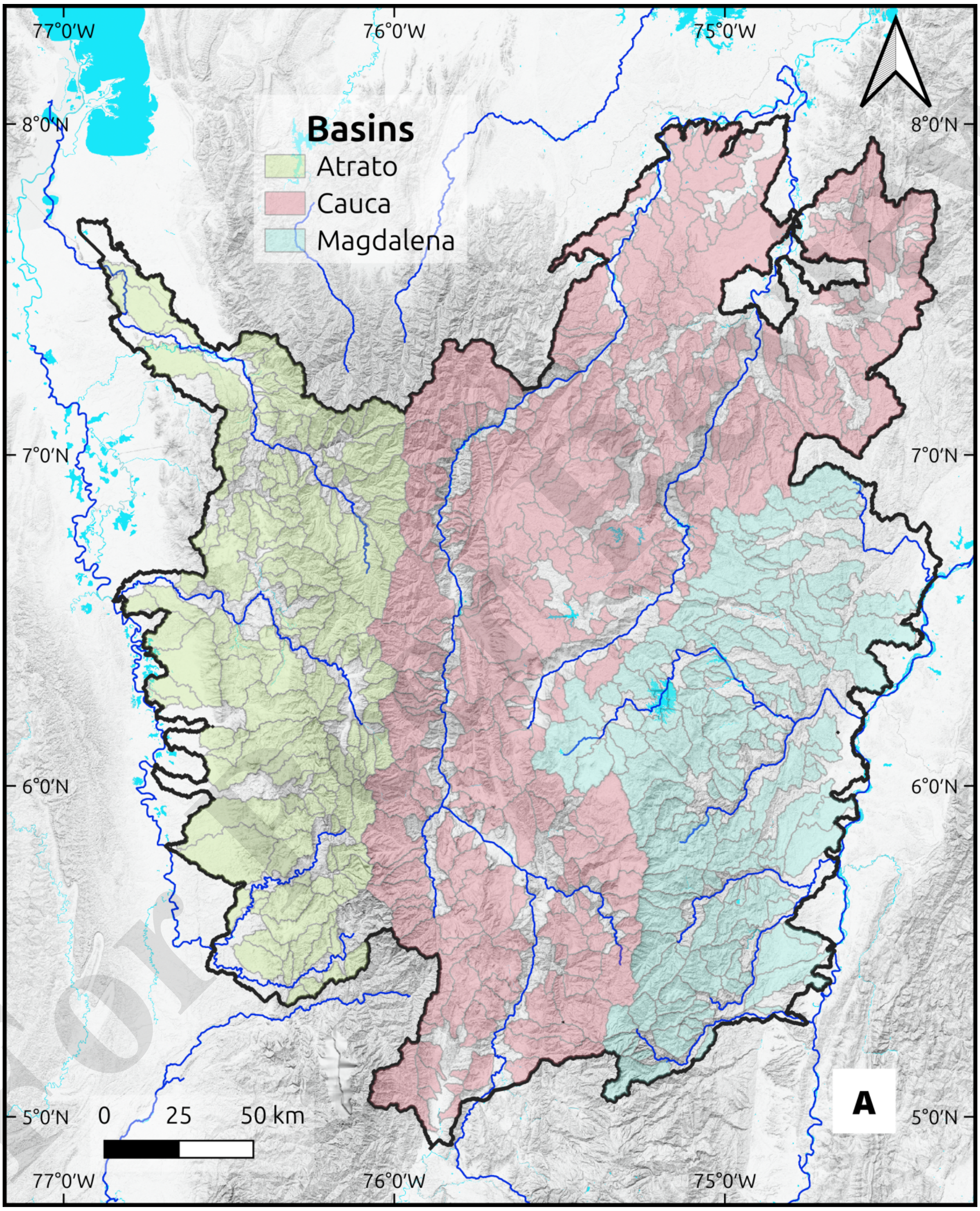


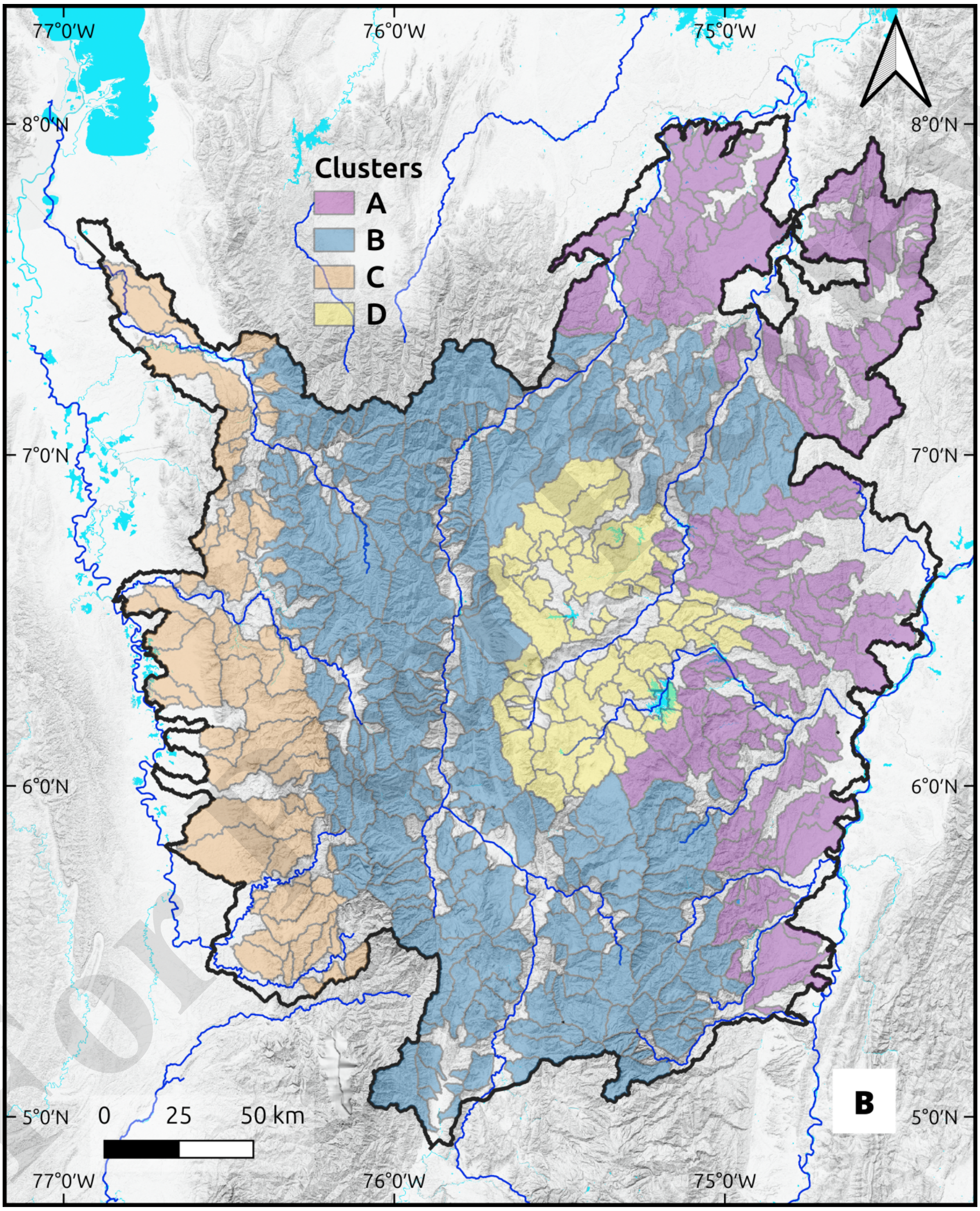
Basin Model
Cluster Model
Coefficient Value
Adj. R^2: 0.5320
AIC: 1578.230
Adj. R^2: 0.5627
AIC: 1546.431
Atrato-Const.
Atrato-Area
Atrato-Elevation
Atrato-Relief
Cauca-Const.
Cauca-Area
Cauca-Elevation
Cauca-Relief
Magdal.-Const.
Magdal.-Area
Magdal.-Elevation
Magdal.-Relief
A-Const.
A-Area
A-Elevat.
A-Relief
B-Const.
B-Area
B-Elevat.
B-Relief
C-Const.
C-Area
C-Elevat.
C-Relief
D-Const.
D-Area
D-Elevat.
D-Relief


N
A
75 km
0.8
0.6
0.4
0.2
Area Coefficient
N
B
75 km
0.52
0.48
0.44
0.40
Elevation Coefficient
N
C
75 km
0.46
0.43
0.40
0.37
Relief Coefficient



77°0'W
76°0'W
75°0'W
8°0'N
7°0'N
6°0'N
5°0'N
Landslides
0 25 50 km
A


77°0'W
76°0'W
75°0'W
8°0'N
7°0'N
6°0'N
5°0'N
Basins
Atrato
Cauca
Magdalena
0
25
50 km
A


77°0'W
76°0'W
75°0'W
8°0'N
7°0'N
6°0'N
5°0'N
Clusters
A
B
C
D
0
25
50 km
B